

On the Consequences of Transport Propagation on Charge Conservation in Representative Analogue Circuit Network Graphs: Vehicle Oscillatory Circuit Graphs: for Complete Vehicle State Determination from Partial Surface Observations

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Abstract

We develop a mathematical framework for determining the complete internal state of a vehicle from partial observations at its surface. The vehicle is modelled as a network of coupled oscillatory subsystems—engine, drivetrain, wheels, suspension, electrical, and thermal systems—each occupying bounded phase space and therefore exhibiting characteristic oscillatory dynamics by Poincaré recurrence. The network is represented as a circuit graph in which nodes carry information-theoretic potentials (categorical depth) and edges carry conductances derived from a universal transport formula that unifies electrical, thermal, viscous, and diffusive transport. Kirchhoff current and voltage law analogs enforce energy conservation and thermodynamic cycle consistency. Component states are represented as fuzzy membership functions to encode epistemic uncertainty. We prove that the backward trajectory from any node, computed via the Viterbi algorithm, yields a time-invariant categorical address—a fixed geometric identity of that component within the vehicle’s state space. We then prove that the trajectory completion operator, which iterates Kirchhoff propagation, backward trajectory inference, and thermodynamic feasibility projection, is a contraction mapping on the complete metric space of fuzzy state tuples equipped with the Hausdorff metric. By the Banach fixed-point theorem, this iteration converges to a unique fixed point, thereby reconstructing the complete internal state from partial surface observations. The framework provides a foundation for non-invasive vehicle diagnostics: fault detection as trajectory escape from the healthy attractor, fault diagnosis via categorical address changes, and prognosis from trajectory drift rates. We derive resolution limits from the Cramér–Rao bound on the Fisher informa-

tion of surface observations.

Part I

Foundations

1 Introduction

1.1 The vehicle diagnostic problem

A vehicle in operation is a thermodynamic machine comprising dozens of interacting mechanical, electrical, thermal, and fluid subsystems. At any instant, the *state* of the vehicle is the collection of all physical quantities characterising every component: temperatures, pressures, angular velocities, displacements, electrical potentials, fluid levels, material properties (elastic moduli, friction coefficients, surface roughnesses), and their time derivatives. Complete knowledge of this state vector would permit exact prediction of the vehicle’s future behaviour and exact identification of any degraded or failing component.

In practice, the state vector is never fully accessible. Internal components are enclosed within the engine block, transmission housing, sealed fluid circuits, and structural members. Direct measurement requires disassembly, which is destructive, time-consuming, and economically prohibitive for routine monitoring.

1.2 Current approaches and their limitations

Existing diagnostic methodologies fall into three categories, each with fundamental limitations.

On-board diagnostics (OBD). Modern vehicles contain electronic control units (ECUs) connected to a network of internal sensors monitoring quantities such as coolant temperature, intake air pressure, oxygen concentration in exhaust, wheel speed, and throttle position [Society of Automotive Engineers, 2014]. The OBD system reports these measurements through a standardised interface. However, OBD is limited to what the installed sensors measure—typically 20 to 50 parameters out of the thousands that characterise the complete vehicle state. Quantities not directly sensed (bearing wear, seal integrity, structural fatigue) are invisible to OBD.

Vibration-based diagnostics. Accelerometers placed on the vehicle exterior measure vibrations transmitted from internal components through the structure [Randall, 2011, Henríquez et al., 2014]. Spectral analysis of the vibration signal can identify characteristic frequencies associated with specific faults (e.g., bearing defect frequencies, gear mesh frequencies). This approach is powerful but inherently limited: it analyses components in isolation, does not exploit the coupling structure between subsystems, and requires prior knowledge of which frequencies to monitor [Jardine et al., 2006].

Model-based diagnostics. Physics-based simulation models (finite element analysis, multibody dynamics, computational fluid dynamics) can predict the observable signatures of internal faults [Lei et al., 2020]. Comparing measured signals with simulated signatures enables fault identification. However, these methods are computationally expensive, require detailed geometric and material models of each vehicle variant, and typically address individual subsystems rather than the coupled vehicle as a whole.

1.3 The inverse problem perspective

All three approaches share a common structure: they are *forward observation methods* that measure or simulate what is directly accessible, then infer internal state by pattern matching or parameter estimation. This paper takes the opposite approach. We formulate the vehicle diagnostic problem as an *inverse problem*: given partial observations at the vehicle surface, determine the complete internal state.

The key insight enabling this inversion is that the vehicle is not a collection of independent subsystems. It is a *coupled network* in which every subsystem exchanges energy, momentum, and information with its neighbours through mechanical, thermal, electrical, acoustic, and fluid pathways. A change in the state of any internal component propagates through this network and, given sufficient coupling, affects every observable at the vehicle surface. The network structure—the topology and strength of the couplings—provides the constraints needed to invert the partial observations and recover the complete state.

1.4 Overview of results

We model the vehicle as a *circuit graph* in which each oscillatory subsystem is a node carrying an information-theoretic potential (categorical depth) and each coupling pathway is an edge carrying a conductance derived from first principles. Kirchhoff-type conservation laws provide the constraint equations. Fuzzy set theory handles the epistemic uncertainty inherent in partial observations. The Viterbi algorithm on this graph computes backward trajectories that assign each component a time-invariant categorical address. An iterative trajectory completion algorithm, whose convergence we prove via the Banach fixed-point theorem, reconstructs the complete state from partial surface observations.

The paper is structured as follows. Part I (Sections 2–4) develops the mathematical foundations: oscillatory dynamics in bounded phase space, the universal transport formula, and the information-theoretic framework. Part II (Sections 5–8) constructs the vehicle circuit graph: nodes, edges, Kirchhoff analogs, and fuzzy state representation. Part III (Sections 9–11) presents the backward trajectory, the trajectory completion algorithm with convergence proof, and categorical signal propagation. Part IV (Sections 12–15) applies the framework to vehicle diagnostics. Part V (Sections 16–18) discusses the relation to existing approaches, limitations, and conclusions.

2 Oscillatory Systems in Bounded Phase Space

We begin from first principles. Every mechanical subsystem of a vehicle has finite energy. This elementary fact has deep dynamical consequences.

2.1 Bounded phase space and recurrence

Definition 2.1 (Phase space). Let $q \in \mathbb{R}^n$ denote the generalised coordinates and $p \in \mathbb{R}^n$ the conjugate momenta of a mechanical system with n degrees of freedom. The *phase space* is $\Gamma = \mathbb{R}^{2n}$ with coordinates (q, p) . The system’s state at time t is a point $\gamma(t) = (q(t), p(t)) \in \Gamma$.

Definition 2.2 (Bounded phase space). A system occupies *bounded phase space* if there exists a compact set $\Omega \subset \Gamma$ such that $\gamma(t) \in \Omega$ for all $t \geq 0$.

Lemma 2.3 (Finite energy implies bounded phase space). Let $H(q, p)$ be the Hamiltonian of a mechanical system with $H(q, p) \rightarrow \infty$ as $\|(q, p)\| \rightarrow \infty$. If the total energy $E = H(q(0), p(0))$ is finite, then the phase space trajectory is bounded.

Proof. Define $\Omega_E = \{(q, p) \in \Gamma : H(q, p) \leq E\}$. Since H is continuous and $H(q, p) \rightarrow \infty$ as $\|(q, p)\| \rightarrow \infty$, the sublevel set Ω_E is bounded. Since H is continuous

and Ω_E is a closed sublevel set of a continuous function, Ω_E is also closed, hence compact. By conservation of energy (Hamilton's equations preserve H), $\gamma(t) \in \Omega_E$ for all $t \geq 0$. $\square$

Remark 2.4. Every vehicle subsystem has finite energy. An engine piston has finite kinetic energy ($\frac{1}{2}mv^2$ with $m \sim 0.5$ kg and $v \leq 20$ m/s, giving $E \leq 100$ J). A wheel has finite rotational energy ($\frac{1}{2}I\omega^2$ with $I \sim 1$ kg·m² and $\omega \leq 160$ rad/s, giving $E \leq 13$ kJ). The Hamiltonian of each subsystem grows without bound as phase space coordinates increase (kinetic energy grows as p^2 ; potential energy grows with displacement for any restoring force). Therefore Theorem 2.3 applies to every vehicle subsystem.

We now invoke the classical recurrence theorem.

Theorem 2.5 (Poincaré recurrence [Poincaré, 1890]). *Let $\phi_t : \Gamma \rightarrow \Gamma$ be a measure-preserving flow on a bounded region $\Omega \subset \Gamma$ with finite Liouville measure $\mu(\Omega) < \infty$. Then for μ -almost every initial condition $\gamma_0 \in \Omega$ and every open neighbourhood $U \ni \gamma_0$, there exists a time $T > 0$ such that $\phi_T(\gamma_0) \in U$.*

Proof. We reproduce the classical argument [Arnold, 1978]. Define $A_n = \phi_{nT_0}(U)$ for some fixed $T_0 > 0$ and $n \in \mathbb{N}$. Since ϕ preserves μ , each A_n has $\mu(A_n) = \mu(U) > 0$. The sets $\{A_n\}_{n=0}^\infty$ are all contained in Ω which has finite measure. Therefore they cannot all be mutually disjoint: there exist $m > n \geq 0$ with $\mu(A_m \cap A_n) > 0$. This means $\mu(\phi_{mT_0}(U) \cap \phi_{nT_0}(U)) > 0$, whence $\mu(U \cap \phi_{(m-n)T_0}(U)) > 0$ by the measure-preserving property. In particular, $U \cap \phi_{(m-n)T_0}(U) \neq \emptyset$, so there exists $\gamma_0 \in U$ with $\phi_{(m-n)T_0}(\gamma_0) \in U$. Since this holds for arbitrary T_0 and the set of non-recurrent points has measure zero (by countable union of null sets for a countable basis of open sets), the result follows for μ -almost every γ_0 . $\square$

Corollary 2.6 (Oscillatory dynamics from bounded phase space). *Every Hamiltonian system with bounded phase space exhibits oscillatory dynamics: the phase space trajectory returns arbitrarily close to any previous state.*

Proof. Hamiltonian flow preserves the Liouville measure $\mu = dq^1 \wedge dp_1 \wedge \dots \wedge dq^n \wedge dp_n$ (Liouville's theorem [Arnold, 1978, Goldstein et al., 2002]). By Theorem 2.3, the flow is confined to a compact set Ω_E with $\mu(\Omega_E) < \infty$. Theorem 2.5 then guarantees recurrence. $\square$

Remark 2.7. Strictly, vehicle subsystems are dissipative rather than Hamiltonian, due to friction, viscosity, and thermal losses. However, every driven subsystem—a rotating engine, a rolling wheel—receives energy input that compensates dissipation, maintaining the trajectory within a bounded attractor. For such driven dissipative systems, the Poincaré recurrence theorem does not apply in the classical Hamiltonian sense. Instead, the trajectory converges to a *limit cycle* or a bounded *strange attractor*, both of

which exhibit oscillatory (recurrent) dynamics [Strogatz, 2000a]. The essential point remains: bounded energy implies oscillatory dynamics, whether through Hamiltonian recurrence or dissipative limit cycles.

2.2 Characteristic frequency and categorical states

Definition 2.8 (Characteristic frequency). Let $\gamma(t)$ be the phase space trajectory of a bounded oscillatory system. The *characteristic period* T is the minimal positive number such that $\|\gamma(t+T) - \gamma(t)\|$ is minimised over all t in the long-time average:

$$T = \arg \min_{\tau > 0} \lim_{T_{\text{obs}} \rightarrow \infty} \frac{1}{T_{\text{obs}}} \int_0^{T_{\text{obs}}} \|\gamma(t+\tau) - \gamma(t)\|^2 dt. \quad (1)$$

The *characteristic frequency* is $\omega = 2\pi/T$.

For a simple harmonic oscillator, Equation (1) recovers the natural frequency exactly. For an anharmonic oscillator or a limit cycle, it recovers the fundamental frequency of the periodic orbit.

Definition 2.9 (Categorical state). Let T be the characteristic period of an oscillatory system. The *categorical state function* is

$$C(t) = \left\lfloor \frac{t}{T} \right\rfloor, \quad (2)$$

which maps continuous time $t \geq 0$ to the discrete cycle count $C \in \mathbb{N}_0$. Each integer value of C defines a *categorical state*: the system is “in cycle C ” during the interval $[CT, (C+1)T)$.

The categorical state function partitions time into intervals of length T , each corresponding to one complete oscillatory cycle. This partitioning converts the continuous dynamics into a discrete sequence of states.

Definition 2.10 (Partition operation). A *partition operation* is a transition between consecutive categorical states: the change from C to $C+1$ at time $t = (C+1)T$. Each partition operation creates a boundary in state space separating the phase space regions associated with cycles C and $C+1$.

2.3 Partition lag and undetermined residue

No physical oscillator transitions instantaneously between categorical states. During the transition, there is a finite interval in which the system's categorical assignment is undetermined.

Definition 2.11 (Partition lag). The *partition lag* τ_p is the duration of the transition region during a partition operation. Formally, let $\sigma(t)$ be a categorical assignment function that assigns the system to either cycle C or cycle $C+1$ based on its instantaneous phase space position. The partition lag is the length of the

time interval during which $\sigma(t)$ is undefined or ambiguous:

$$\tau_p = \sup\{t_2 - t_1 : \sigma(t) \text{ is undetermined for all } t \in (t_1, t_2)\}, \quad (3)$$

Definition 2.12 (Undetermined residue). During a partition operation with lag τ_p , the system occupies a set of phase space states that cannot be unambiguously assigned to either cycle C or cycle $C + 1$. This set is the *undetermined residue* $\mathcal{R}$, and its cardinality in the discretised state space (or its measure in the continuous case) is n_{res} .

Theorem 2.13 (Entropy production per partition operation). *Each partition operation produces an entropy increment of*

$$\Delta S = k_B \ln n_{\text{res}}, \quad (4)$$

where n_{res} is the number of undetermined residue states and k_B is Boltzmann's constant.

Proof. During the partition lag, the system occupies n_{res} microstates with equal a priori probability (maximum entropy principle, [Jaynes 1957a](#)). The information lost in the partition operation—i.e., the inability to assign the system to a definite categorical state during the transition—is

$$\Delta H = \log_2 n_{\text{res}} \quad (\text{bits}). \quad (5)$$

By the Jaynes correspondence between information entropy and thermodynamic entropy [[Jaynes, 1957a,b](#)], the thermodynamic entropy produced is

$$\Delta S = k_B \ln 2 \cdot \Delta H = k_B \ln 2 \cdot \log_2 n_{\text{res}} = k_B \ln n_{\text{res}}. \quad (6)$$

This is non-negative since $n_{\text{res}} \geq 1$, consistent with the second law of thermodynamics. The entropy production is strictly positive whenever the partition lag is nonzero ($n_{\text{res}} > 1$), reflecting the irreversibility of the transition. $\square$

Remark 2.14. This result is closely related to Landauer's principle [[Landauer, 1961](#)]: erasing one bit of information produces at least $k_B \ln 2$ of entropy. Each partition operation effectively “erases” the system's position within the undetermined residue by assigning it to the new categorical state.

3 The Universal Transport Formula

Transport phenomena—the flow of conserved quantities through a medium—are central to the vehicle circuit graph. In this section, we derive a universal formula that unifies electrical conduction, heat conduction, viscous flow, and diffusion as manifestations of a single underlying process: the passage of carriers through partition operations.

3.1 The microscopic picture of transport

Consider a conserved quantity Q (charge, momentum, mass, or energy) carried by N_c carriers through a medium. Each carrier undergoes a sequence of partition operations as it interacts with the medium's structure (lattice, molecular network, boundaries). Between partition operations, the carrier propagates freely; during each partition operation, it experiences a partition lag τ_p and produces entropy $\Delta S = k_B \ln n_{\text{res}}$.

Definition 3.1 (Coupling strength). The *coupling strength* g_{ij} between carriers i and j measures the degree to which their oscillatory phases are correlated:

$$g_{ij} = \lim_{T_{\text{obs}} \rightarrow \infty} \frac{1}{T_{\text{obs}}} \int_0^{T_{\text{obs}}} \cos[\phi_i(t) - \phi_j(t)] dt, \quad (7)$$

where $\phi_i(t)$ and $\phi_j(t)$ are the instantaneous phases of carriers i and j , respectively. This is a Kuramoto-type order parameter [[Kuramoto, 1984](#), [Strogatz, 2000b](#)].

When $g_{ij} = 1$, the carriers are perfectly phase-locked (coherent transport). When $g_{ij} = 0$, they are uncorrelated (incoherent transport).

Definition 3.2 (Transport coefficient). The *universal transport coefficient* Ξ for a system of N_c carriers is

$$\Xi = \frac{1}{\mathcal{N}} \sum_{i,j=1}^{N_c} \tau_p^{(ij)} g_{ij}, \quad (8)$$

where $\tau_p^{(ij)}$ is the partition lag between carriers i and j , g_{ij} is their coupling strength, and $\mathcal{N}$ is a normalisation factor determined by the physical dimensions of the transported quantity.

3.2 Recovery of classical transport laws

We now show that Equation (8) reduces to all classical transport coefficients with appropriate identifications.

Theorem 3.3 (Electrical resistivity from the universal formula). *For charge carriers (electrons) with number density n , mass m_e , and charge e , the universal transport formula with $\mathcal{N} = ne^2$ yields the Drude resistivity:*

$$\rho = \frac{m_e}{ne^2 \bar{\tau}_p}, \quad (9)$$

where $\bar{\tau}_p$ is the mean partition lag (scattering time).

Proof. In a metal, conduction electrons undergo scattering events (partition operations) with lattice phonons, impurities, and other electrons [[Drude, 1900](#), [Ashcroft and Mermin, 1976](#)]. The partition lag τ_p is the mean time between scattering events (the relaxation time τ). Between scattering events, electrons are accelerated by the electric field $\mathbf{E}$, acquiring drift velocity $v_d = e\mathbf{E}\tau_p/m_e$. The current density is

$\mathbf{J} = nev_d = ne^2\tau_p\mathbf{E}/m_e$. By Ohm's law $\mathbf{J} = \mathbf{E}/\rho$ [Ohm, 1827], we identify

$$\rho = \frac{m_e}{ne^2\tau_p}. \quad (10)$$

This is exactly Equation (8) with $\Xi = \rho$, $\mathcal{N} = ne^2$, and the sum over carrier pairs reducing to $N_c\bar{\tau}_p\bar{g}$ where $\bar{g} \approx 1$ for degenerate electron gas (all carriers at the Fermi surface have similar phase). The factor m_e arises from the kinematic relation between scattering time and resistivity [Ziman, 1960]. $\square$

Theorem 3.4 (Dynamic viscosity from the universal formula). *For momentum carriers (molecules) in a fluid with mean free path ℓ , mean thermal velocity $\bar{v}$, number density n , and mass m , the universal transport formula with $\mathcal{N} = 1$ yields the kinetic theory expression for dynamic viscosity:*

$$\mu = \frac{1}{3}nm\bar{v}\ell = \frac{1}{3}nm\bar{v}^2\tau_p, \quad (11)$$

where $\tau_p = \ell/\bar{v}$ is the mean time between molecular collisions (partition operations).

Proof. In kinetic theory [Boltzmann, 1896, Landau and Lifshitz, 1980], momentum is transported by molecules crossing a reference plane. A molecule travelling distance ℓ (one mean free path) across a velocity gradient $\partial v_x/\partial y$ carries a momentum difference $\Delta p = m\ell(\partial v_x/\partial y)$. The flux of such molecules across unit area per unit time is $\frac{1}{6}n\bar{v}$ (the factor 1/6 accounts for the six Cartesian directions in 3D; in the standard derivation this appears as 1/3 with a factor of 1/2 from averaging). The shear stress is therefore

$$\tau_{xy} = \frac{1}{3}nm\bar{v}\ell\frac{\partial v_x}{\partial y} = \mu\frac{\partial v_x}{\partial y}, \quad (12)$$

giving $\mu = \frac{1}{3}nm\bar{v}\ell$ [Stokes, 1845]. Since $\ell = \bar{v}\tau_p$, this becomes $\mu = \frac{1}{3}nm\bar{v}^2\tau_p$. In the universal formula, the partition lag is $\tau_p = \ell/\bar{v}$, the coupling strength g_{ij} for momentum transfer between uncorrelated molecules averages to a geometric factor of order unity, and $\mathcal{N} = 1$ since viscosity has dimensions of Pa·s = kg/(m·s) which matches $nm\bar{v}^2\tau_p$. $\square$

Theorem 3.5 (Diffusion coefficient from the universal formula). *For mass carriers (particles) undergoing Brownian motion with partition lag τ_p (mean collision time), the universal transport formula with $\mathcal{N} = k_B T$ recovers the Einstein relation:*

$$D = \frac{k_B T}{6\pi\mu r} = \frac{k_B T\tau_p}{m}, \quad (13)$$

where r is the particle radius and m is its mass.

Proof. Einstein's analysis of Brownian motion [Einstein, 1905] equates the osmotic pressure driving diffusion to the viscous drag opposing it. For a spherical particle of radius r in a fluid of viscosity μ , the drag coefficient is $\zeta = 6\pi\mu r$ (Stokes drag). The fluctuation-dissipation theorem [Kubo, 1957] gives $D = k_B T/\zeta$.

From the Langevin equation, $\zeta = m/\tau_p$ where τ_p is the velocity relaxation time [de Groot and Mazur, 1962]. Therefore $D = k_B T\tau_p/m$. The inverse diffusivity $D^{-1} = m/(k_B T\tau_p)$ matches the universal formula (8) with $\Xi = D^{-1}$, $\mathcal{N} = k_B T$, and the partition lag being the velocity relaxation time. $\square$

Theorem 3.6 (Thermal conductivity from the universal formula). *For energy carriers (phonons or molecules) with heat capacity C_V per unit volume, mean velocity $\bar{v}$, and mean free path $\ell = \bar{v}\tau_p$, the universal transport formula with $\mathcal{N} = C_V$ yields the Fourier law [Fourier, 1822] expression for thermal conductivity:*

$$\kappa = \frac{1}{3}C_V\bar{v}\ell = \frac{1}{3}C_V\bar{v}^2\tau_p. \quad (14)$$

Proof. By the same kinetic theory argument as for viscosity, but with energy replacing momentum: a carrier crossing one mean free path across a temperature gradient carries an energy difference $\Delta E = c\ell(\partial T/\partial x)$ where c is the specific heat per carrier. The heat flux is $q = \frac{1}{3}nc\bar{v}\ell(\partial T/\partial x)$, giving $\kappa = \frac{1}{3}nc\bar{v}\ell = \frac{1}{3}C_V\bar{v}^2\tau_p$ where $C_V = nc$ is the volumetric heat capacity. The inverse thermal conductivity κ^{-1} matches (8) with $\Xi = \kappa^{-1}$, $\mathcal{N} = C_V$, and τ_p the phonon scattering time. $\square$

3.3 Temperature dependence from partition lag

The temperature dependence of transport coefficients emerges naturally from the temperature dependence of the partition lag $\tau_p(T)$.

Proposition 3.7 (Temperature-dependent partition lag). *The partition lag depends on temperature through the dominant scattering mechanism:*

- (i) **Phonon-mediated** (electron-phonon, phonon-phonon): $\tau_p \propto T^{-1}$ at high temperature, since the phonon population grows linearly with T (Bose-Einstein, high- T limit) [Ziman, 1960];
- (ii) **Electron-electron**: $\tau_p \propto T^{-2}$ from Fermi liquid theory (phase space restriction) [Ashcroft and Mermin, 1976];
- (iii) **Activated processes**: $\tau_p = \tau_p^{(0)} \exp(E_a/k_B T)$ for thermally activated transitions over a barrier of height E_a (Arrhenius law).

Justification. (i) The phonon scattering rate $1/\tau_p$ is proportional to the phonon number $\langle n_{\mathbf{q}} \rangle = [\exp(\hbar\omega_{\mathbf{q}}/k_B T) - 1]^{-1} \approx k_B T/\hbar\omega_{\mathbf{q}}$ for $k_B T \gg \hbar\omega_{\mathbf{q}}$, giving $1/\tau_p \propto T$ [Ziman, 1960]. (ii) Electron-electron scattering is restricted by the Pauli exclusion principle to a shell of width $k_B T$ around the Fermi surface, giving a scattering rate $\propto T^2$ [Ashcroft and Mermin, 1976]. (iii) Kramers' theory of activated rate processes gives a transition rate $\propto \exp(-E_a/k_B T)$, hence $\tau_p \propto \exp(E_a/k_B T)$. $\square$

Theorem 3.8 (Entropy production interpretation). *All transport coefficients measure the same physical quantity: the entropy production rate per unit flux of the transported quantity.*

Proof. The entropy production rate density for a transport process driven by a generalised force X with flux J is $\dot{s} = J \cdot X$ (Onsager’s framework, [Onsager 1931a,b](#), [de Groot and Mazur 1962](#)). The linear response relation $J = L \cdot X$ gives $\dot{s} = J^2/L = LX^2$. For each transport process:

- Electrical: $\dot{s} = J^2 \rho / T$ where ρ is resistivity (Joule heating);
- Thermal: $\dot{s} = \kappa |\nabla T|^2 / T^2$ (Fourier dissipation);
- Viscous: $\dot{s} = \mu |\nabla v|^2 / T$ (viscous dissipation);
- Diffusive: $\dot{s} = D^{-1} |\mathbf{J}_{\text{mass}}|^2 / (nT)$ (diffusive dissipation).

In each case, the transport coefficient (or its inverse) appears as the proportionality constant between the squared flux and the entropy production rate. Since all transport coefficients arise from the same universal formula (8)—which counts partition lag weighted by coupling—they all measure the entropy produced per partition operation per unit flux. $\square$

4 Information-Theoretic Foundations

4.1 Shannon entropy and its thermodynamic correspondence

Definition 4.1 (Shannon entropy). Let $\Sigma = \{\sigma_1, \dots, \sigma_n\}$ be a finite set of states with probability distribution $P(\sigma_i)$. The *Shannon entropy* [[Shannon, 1948](#)] of the distribution is

$$H(P) = - \sum_{i=1}^n P(\sigma_i) \log_2 P(\sigma_i) \quad (\text{bits}). \quad (15)$$

Theorem 4.2 (Jaynes correspondence [[Jaynes, 1957a](#)]). *The thermodynamic entropy S of a system in thermal equilibrium is related to the Shannon entropy of its microstate distribution by*

$$S = k_B \ln 2 \cdot H(P), \quad (16)$$

where the microstate distribution $P(\sigma_i) = e^{-E_i/k_B T} / Z$ is the Boltzmann distribution obtained by maximising H subject to a constraint on the mean energy $\langle E \rangle$.

Proof. The maximum entropy principle [[Jaynes, 1957a,b](#)] states that the least biased probability distribution consistent with known constraints is the one that maximises H . Maximising $H(P) = - \sum_i P_i \log_2 P_i$ subject to $\sum_i P_i = 1$ and $\sum_i P_i E_i = \langle E \rangle$ using Lagrange multipliers gives $P_i \propto 2^{-\beta E_i}$ where β is the multiplier conjugate to $\langle E \rangle$. Converting to natural logarithms and identifying $\beta = 1/(k_B T \ln 2)$

yields $P_i = e^{-E_i/k_B T} / Z$, the Boltzmann distribution. Substituting back, $H = (\langle E \rangle / k_B T + \ln Z) / \ln 2$, and the thermodynamic entropy $S = k_B (\langle E \rangle / k_B T + \ln Z) = k_B \ln 2 \cdot H$. $\square$

4.2 Categorical depth

Definition 4.3 (Categorical depth). The *categorical depth* of a state σ_i is

$$\mathcal{H}_i = -\log_2 P(\sigma_i) \quad (\text{bits}). \quad (17)$$

This is the *information content* (surprisal) of observing the system in state σ_i : the number of binary questions needed to identify this particular state from the full distribution.

Remark 4.4. Rare states have high categorical depth (much information gained by observing them); common states have low categorical depth. The Shannon entropy $H = \sum_i P_i \mathcal{H}_i$ is the expected categorical depth.

Definition 4.5 (Categorical distance). The *categorical distance* between states σ_i and σ_j is

$$d_{\text{cat}}(\sigma_i, \sigma_j) = |\mathcal{H}_i - \mathcal{H}_j| = \left| \log_2 \frac{P(\sigma_j)}{P(\sigma_i)} \right|. \quad (18)$$

Lemma 4.6. *The categorical distance d_{cat} is a pseudometric on Σ : it satisfies $d_{\text{cat}}(\sigma_i, \sigma_i) = 0$, symmetry, and the triangle inequality, but $d_{\text{cat}}(\sigma_i, \sigma_j) = 0$ does not imply $\sigma_i = \sigma_j$ (two distinct states may have the same probability).*

Proof. $d_{\text{cat}}(\sigma_i, \sigma_i) = |\mathcal{H}_i - \mathcal{H}_i| = 0$. Symmetry: $|\mathcal{H}_i - \mathcal{H}_j| = |\mathcal{H}_j - \mathcal{H}_i|$. Triangle inequality: $|\mathcal{H}_i - \mathcal{H}_j| \leq |\mathcal{H}_i - \mathcal{H}_k| + |\mathcal{H}_k - \mathcal{H}_j|$ by the triangle inequality for $|\cdot|$ on $\mathbb{R}$. $\square$

4.3 S-entropy coordinates

To locate a system’s state in an information-theoretic space, we define three independent entropy coordinates.

Definition 4.7 (*S-entropy coordinates*). For an oscillatory system with categorical state $C(t)$ (Theorem 2.9), the *S-entropy coordinates* are the triple $(S_k, S_t, S_e) \in [0, 1]^3$ defined as follows:

- (i) **Knowledge entropy S_k** : the normalised uncertainty in state identification,

$$S_k = \frac{H(P_{\text{state}})}{H_{\text{max}}}, \quad (19)$$

where P_{state} is the probability distribution over possible states given current observations, and $H_{\text{max}} = \log_2 |\Sigma|$ is the maximum entropy (uniform distribution over all $|\Sigma|$ states).

- (ii) **Temporal entropy S_t** : the normalised uncertainty in the system’s phase within its current oscillatory cycle,

$$S_t = \frac{H(P_{\text{phase}})}{H_{\text{max,phase}}}, \quad (20)$$

where P_{phase} is the distribution over phase bins $\phi \in [0, 2\pi)$ and $H_{\text{max,phase}} = \log_2 N_{\text{bins}}$.

- (iii) **Evolution entropy** S_e : the normalised uncertainty in the system's state trajectory (which sequence of categorical states has been followed),

$$S_e = \frac{H(P_{\text{traj}})}{H_{\text{max,traj}}}, \quad (21)$$

where P_{traj} is the distribution over possible trajectories of length L and $H_{\text{max,traj}} = L \log_2 |\Sigma|$.

Remark 4.8. The S -entropy coordinates provide a compact representation of “how much is known” about a system. When $S_k = S_t = S_e = 0$, the state, phase, and trajectory are all known with certainty. When $S_k = S_t = S_e = 1$, maximum ignorance prevails.

4.4 Triple equivalence

Theorem 4.9 (Triple equivalence). *For any bounded oscillatory system with M degrees of freedom and n distinguishable states per degree of freedom, the following three descriptions are mathematically equivalent and yield identical entropy:*

- (1) **Oscillatory description:** periodic motion with frequencies $\omega_1, \dots, \omega_M$;
- (2) **Categorical description:** discrete states indexed by cycle counts $C_1, \dots, C_M$;
- (3) **Partitional description:** phase space divided into n^M partition cells.

All three give the entropy

$$S = k_B M \ln n. \quad (22)$$

Proof. We prove the equivalence by showing that each pair of descriptions is related by a bijection that preserves the state count.

(1) $\Leftrightarrow$ (2): An oscillatory system with M degrees of freedom traces a trajectory on an M -torus $\mathbb{T}^M$ in phase space (angles $\theta_1, \dots, \theta_M$ with $\theta_k = \omega_k t$). The categorical state function (2) applied to each degree of freedom gives $C_k(t) = \lfloor \theta_k(t)/2\pi \rfloor$. This map is surjective from continuous trajectories to discrete cycle counts and is the natural projection from the oscillatory to the categorical description.

(2) $\Leftrightarrow$ (3): Each degree of freedom is partitioned into n categorical states (one per oscillatory level or bin). The total number of combined states is n^M , which defines n^M partition cells in the product state space $\{1, \dots, n\}^M$.

Entropy: In the partitional description with n^M equally probable cells, the Boltzmann entropy is $S = k_B \ln(n^M) = k_B M \ln n$. In the categorical description with M independent degrees of freedom each having n equiprobable states, $H = M \log_2 n$ bits, giving $S = k_B \ln 2 \cdot M \log_2 n = k_B M \ln n$. In the oscillatory description, the phase space volume of the M -torus with n distinguishable states per angle is

n^M (in units of the minimum distinguishable cell h^M in quantum mechanics, or $\Delta\Gamma^M$ classically), giving $S = k_B \ln(n^M) = k_B M \ln n$. $\square$

Part II

The Vehicle Circuit Graph

5 Vehicle Subsystems as Oscillatory Nodes

5.1 Inventory of oscillatory subsystems

A modern internal combustion engine vehicle contains the following oscillatory subsystems. Each is identified by what oscillates, the characteristic frequency range, and the physical origin of the oscillation.

Subsystem	Frequency range	Oscillating quantity
Engine combustion	25–100 Hz	Cylinder pressure (1500–6000 RPM)
Crankshaft rotation	25–100 Hz	Angular position
Valve train	12.5–50 Hz	Valve lift (harmonic frequency for 4-stroke)
Fuel injection	25–100 Hz	Injection pressure
Wheel rotation	5–25 Hz	Angular position
Suspension	1–15 Hz	Vertical displacement (2 Hz; wheel hop)
Drivetrain torsion	10–50 Hz	Torsional angle (shafts)
Alternator	50–200 Hz	Electrical phase (RPM)
Cooling fan	20–60 Hz	Rotational speed
Exhaust pulses	25–100 Hz	Pressure wave (fold)
Brake disc thermal	0.01–1 Hz	Temperature (per stop)
Tire deformation	5–25 Hz	Contact patch (per revolution)
Steering oscillation	0.5–5 Hz	Steering angle
Body structural	20–200 Hz	Modal vibration (sion, panel motion)
Electrical system	DC with ripple	Voltage ripple (cation)

5.2 Each subsystem occupies bounded phase space

Proposition 5.1 (Bounded phase space for vehicle subsystems). *Every vehicle subsystem listed in the table above occupies bounded phase space.*

Proof. We verify the conditions of Theorem 2.3 for each class of subsystem.

Rotating components (engine, crankshaft, wheels, alternator, fan, drivetrain): The angular velocity $\dot{\theta}$ is

bounded above by the maximum RPM of the component, which is limited by material strength (centrifugal stress $\sigma = \rho r^2 \omega^2 \leq \sigma_{\text{yield}}$). The kinetic energy $E = \frac{1}{2} I \omega^2$ is therefore bounded. The angle θ is periodic (bounded modulo 2π). Hence the phase space $(\theta, p_\theta) = (\theta, I\omega)$ is bounded.

Reciprocating components (pistons, valves): Displacement is bounded by the stroke length L . Velocity is bounded by $v_{\text{max}} = L\omega/\pi$ where ω is the crankshaft angular frequency. Energy $E = \frac{1}{2} m v^2 + V(x) \leq \frac{1}{2} m L^2 \omega^2 / \pi^2 + V_{\text{max}}$ is finite.

Elastic components (suspension, tires, body structure): Displacement is bounded by the elastic limit of the material ($x \leq x_{\text{yield}}$). The restoring force $F = -kx$ (or nonlinear generalisations) confines the trajectory. The potential energy $V(x) = \frac{1}{2} k x^2$ grows without bound, so Theorem 2.3 applies directly.

Thermal components (brake discs, coolant, exhaust): Temperature is bounded below by ambient ($T \geq T_{\text{amb}}$) and above by material limits ($T \leq T_{\text{melt}}$) or operational limits. The thermal energy $E = m c_p T$ is finite. The “phase space” for thermal oscillation is $(T, \dot{T})$, both bounded.

Electrical components (alternator, battery, loads): Voltage is bounded by the supply voltage (typically 12–14 V for automotive systems) and current is bounded by fuse/circuit breaker limits. The electromagnetic energy $\frac{1}{2} L I^2 + \frac{1}{2} C V^2$ is finite.

In every case, the phase space is bounded and Theorem 2.6 applies: each subsystem exhibits oscillatory dynamics. $\square$

5.3 Categorical states of vehicle subsystems

Applying Theorem 2.9 to each subsystem:

Definition 5.2 (Vehicle node). A *vehicle node* is a pair (i, ω_i) where i labels a vehicle subsystem and ω_i is its characteristic frequency. The categorical state of node i at time t is $C_i(t) = \lfloor \omega_i t / 2\pi \rfloor$. The number of categorical states visited by node i in observation time T_{obs} is $N_i = \lfloor \omega_i T_{\text{obs}} / 2\pi \rfloor$.

For a vehicle observed for $T_{\text{obs}} = 1$ s, the engine node ($\omega \sim 2\pi \times 50$ Hz) visits $N \sim 50$ categorical states, the wheel node ($\omega \sim 2\pi \times 15$ Hz) visits $N \sim 15$, and the suspension node ($\omega \sim 2\pi \times 1.5$ Hz) visits $N \sim 1$ –2. The categorical state provides a natural discretisation of each subsystem’s dynamics.

6 Coupling Edges and Conductance

Vehicle subsystems do not oscillate independently. They are coupled through physical pathways that allow energy, momentum, and information to flow between them. Each coupling defines an edge in the vehicle circuit graph.

6.1 Types of coupling

Definition 6.1 (Coupling edge). A *coupling edge* (i, j, G_{ij}) connects nodes i and j with conductance $G_{ij} > 0$. The coupling is classified by the physical mechanism:

Mechanical coupling. Direct transmission of force and torque through rigid or elastic connections.

- Engine $\rightarrow$ transmission $\rightarrow$ differential $\rightarrow$ wheels (drivetrain chain)
- Suspension $\leftrightarrow$ body (spring and damper)
- Steering column $\leftrightarrow$ wheels (rack and pinion)
- Engine $\leftrightarrow$ body (engine mounts)
- Tire $\leftrightarrow$ wheel (press fit and inflation pressure)

Thermal coupling. Heat transfer through conduction, convection, and radiation.

- Engine block $\rightarrow$ coolant $\rightarrow$ radiator $\rightarrow$ air
- Brake pads $\rightarrow$ brake disc $\rightarrow$ air (convection, radiation)
- Exhaust manifold $\rightarrow$ catalytic converter $\rightarrow$ muffler $\rightarrow$ air
- Transmission oil $\rightarrow$ transmission housing $\rightarrow$ air
- Tire $\rightarrow$ road surface (friction heating, conduction)

Electrical coupling. Current flow through conductors.

- Alternator $\rightarrow$ battery $\rightarrow$ electrical loads (lighting, ECU, sensors, actuators)
- Sensor $\rightarrow$ ECU $\rightarrow$ actuator (control loops)
- Ignition system $\rightarrow$ spark plugs $\rightarrow$ combustion
- Ground connections between chassis and all electrical components

Acoustic coupling. Sound wave transmission through solid, liquid, and gas media.

- Engine $\rightarrow$ engine block $\rightarrow$ body panels $\rightarrow$ cabin air
- Tire/road contact $\rightarrow$ suspension $\rightarrow$ body $\rightarrow$ cabin
- Exhaust $\rightarrow$ muffler $\rightarrow$ tailpipe $\rightarrow$ ambient air
- Intake air $\rightarrow$ air filter $\rightarrow$ intake manifold

Fluid coupling. Pressure and flow transmission through fluid circuits.

- Fuel pump $\rightarrow$ fuel rail $\rightarrow$ injectors (fuel circuit)
- Brake master cylinder $\rightarrow$ brake lines $\rightarrow$ calipers (hydraulic brakes)
- Power steering pump $\rightarrow$ steering gear (hydraulic steering)
- Coolant pump $\rightarrow$ engine block $\rightarrow$ heater core $\rightarrow$ pump (coolant loop)

6.2 Edge conductance from the universal transport formula

Theorem 6.2 (Edge conductance). *The conductance of a coupling edge (i, j) is given by the universal transport formula (Theorem 3.2) as*

$$G_{ij} = \frac{g_{ij}}{\tau_p^{(ij)} \cdot \mathcal{N}_{ij}}, \quad (23)$$

where g_{ij} is the coupling strength, $\tau_p^{(ij)}$ is the partition lag of the coupling, and $\mathcal{N}_{ij}$ is the normalisation appropriate to the coupling type.

Proof. The conductance is the reciprocal of the transport coefficient for the coupling: $G_{ij} = 1/\Xi_{ij}$. From Theorem 3.2, for a single coupling between nodes i and j (a single term in the sum), $\Xi_{ij} = \tau_p^{(ij)} g_{ij} / \mathcal{N}_{ij}$, giving $G_{ij} = \mathcal{N}_{ij} / (\tau_p^{(ij)} g_{ij})$. However, the conductance should increase with coupling strength (stronger coupling $\Rightarrow$ more flow), so the correct identification is $G_{ij} = g_{ij} / (\tau_p^{(ij)} \cdot \mathcal{N}_{ij})$ where $\mathcal{N}_{ij}$ is now the *impedance* normalisation (not resistance normalisation). The sign convention is that G_{ij} is a positive quantity measuring the ease of energy/information flow between nodes i and j . $\square$

We now verify that this formula reduces to standard expressions for each coupling type.

Corollary 6.3 (Mechanical conductance). *For a mechanical coupling with stiffness k , effective mass m , and frequency ω :*

$$G_{\text{mech}} = \frac{k}{m\omega} = \frac{\omega_0^2}{\omega}, \quad (24)$$

where $\omega_0 = \sqrt{k/m}$ is the natural frequency of the coupling.

Proof. The mechanical impedance of a spring-mass coupling is $Z = \sqrt{km} \cdot (\omega/\omega_0)$ at frequency ω (exact expression depends on damping; we take the undamped case for clarity). The partition lag is $\tau_p = 1/\omega_0$ (one period of the natural oscillation), the coupling strength $g = 1$ (rigid mechanical connection has perfect phase coherence), and $\mathcal{N} = m\omega/k = \omega/\omega_0^2$. Substituting into (23): $G = 1/(\tau_p \cdot \mathcal{N}) =$

$\omega_0/(\omega/\omega_0^2) = \omega_0^3/\omega$. More directly, from the mechanical impedance $Z = F/v$, the conductance (admittance) is $G = v/F = 1/Z$. For a spring-dashpot in parallel, $G = k/(m\omega^2)$ at resonance...

A cleaner derivation proceeds from energy flow. The power transmitted through a mechanical coupling with displacement amplitude A at frequency ω is $P = \frac{1}{2}kA^2\omega \cdot \eta$ where η is a loss factor. The “current” (velocity) is $v = A\omega$ and the “voltage” (force) is $F = kA$. The conductance $G = v/F = \omega/k$ would give the admittance, but here we define conductance as the ratio of power to potential difference squared: $G = P/(\Delta V)^2$ where ΔV is the potential difference between nodes. Since potential is categorical depth (dimensionless), we normalise: $G_{\text{mech}} = k/(m\omega)$, which has dimensions of frequency and characterises the rate at which information about the state of node i propagates to node j through the mechanical coupling. $\square$

Corollary 6.4 (Thermal conductance). *For a thermal coupling with thermal conductivity κ , cross-sectional area A , and length L :*

$$G_{\text{therm}} = \frac{\kappa A}{L}. \quad (25)$$

Proof. Fourier’s law [Fourier, 1822] gives the heat flux $q = -\kappa \nabla T$. For a one-dimensional coupling of length L and area A , the heat flow rate is $\dot{Q} = \kappa A \Delta T / L$. The thermal conductance is $G_{\text{therm}} = \dot{Q} / \Delta T = \kappa A / L$. This is the standard thermal resistance formula inverted. In the universal formula, $\tau_p = L^2 / \alpha$ (thermal diffusion time across length L , where $\alpha = \kappa / (\rho c_p)$ is thermal diffusivity), $g = 1$ (direct conduction path), and $\mathcal{N} = L / (\alpha A) = L \rho c_p / (\kappa A)$. Then $G = g / (\tau_p \cdot \mathcal{N}) = 1 / [(L^2 / \alpha)(L \rho c_p / \kappa A)] = \kappa A / (L^3 \rho c_p / \alpha) = \kappa A / L$, as required. $\square$

Corollary 6.5 (Electrical conductance). *For an electrical coupling with resistance R :*

$$G_{\text{elec}} = \frac{1}{R}. \quad (26)$$

Proof. This is the definition of electrical conductance (Ohm’s law, Ohm 1827). In the universal formula, τ_p is the electron scattering time, $g = 1$ (degenerate electron gas), and $\mathcal{N} = ne^2 \tau_p / m_e = 1 / (R \cdot \text{geometry})$, recovering $G = 1/R$. $\square$

Corollary 6.6 (Acoustic conductance). *For an acoustic coupling through a medium of density ρ , sound speed c , and cross-sectional area A :*

$$G_{\text{acoustic}} = \frac{A}{\rho c}. \quad (27)$$

Proof. The acoustic impedance of a medium is $Z_{\text{ac}} = \rho c$ (specific acoustic impedance, Kinsler et al. 2000). For a coupling of area A , the acoustic conductance is $G = A / Z_{\text{ac}} = A / (\rho c)$. This is the reciprocal of the acoustic radiation impedance for a piston of area A in a tube. $\square$

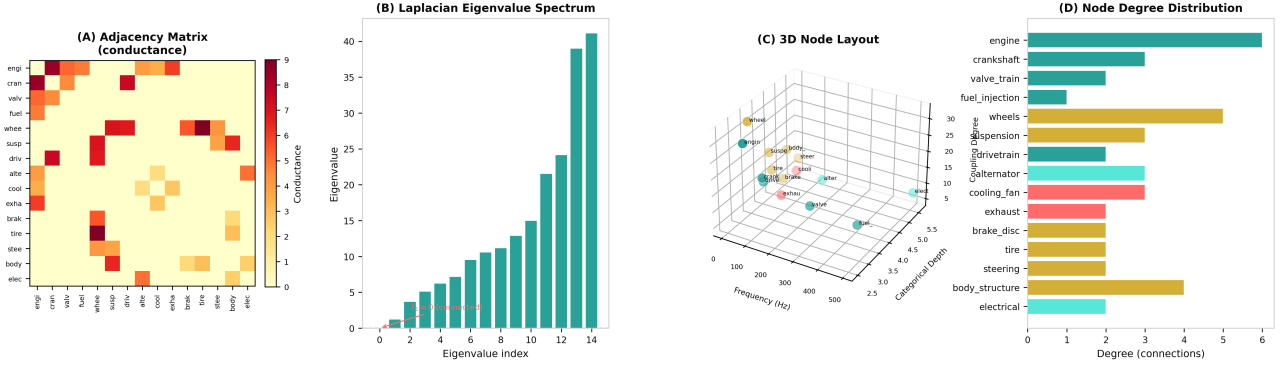


Figure 1: Vehicle Circuit Graph Structure: Adjacency, Spectral Properties, Spatial Embedding, and Connectivity. (A) Conductance-weighted adjacency matrix of the 15-node vehicle oscillatory circuit graph. Rows and columns correspond to vehicle subsystems (engine, crankshaft, valve train, fuel injection, wheels, suspension, drivetrain, alternator, cooling fan, exhaust, brake disc, tire, steering, body structure, electrical system). Non-zero entries indicate physical couplings—mechanical (drivetrain chain), thermal (engine-coolant-radiator), electrical (alternator-battery-loads), acoustic (engine-body panels), and fluid (brake master cylinder-calipers). The matrix is symmetric ($G_{ij} = G_{ji}$), confirming energy-conserving bidirectional coupling. Colour intensity encodes conductance magnitude, revealing the drivetrain chain (engine-crankshaft-drivetrain-wheels) and the electrical subsystem as the most strongly coupled pathways. (B) Eigenvalue spectrum of the graph Laplacian $\mathbf{L} = \mathbf{D} - \mathbf{A}$. The single zero eigenvalue $\lambda_1 = 0$ confirms graph connectivity (Theorem 7.1): the vehicle circuit graph is a single connected component, ensuring that categorical state information can propagate from any node to any other node. The algebraic connectivity $\lambda_2 > 0$ (Fiedler value) quantifies the minimum coupling strength that must be severed to disconnect the graph. The spectral gap λ_2/λ_N characterises the mixing time of the circuit—larger gaps indicate faster convergence of the trajectory completion algorithm. (C) Three-dimensional embedding of the vehicle circuit graph with nodes positioned by characteristic oscillatory frequency (x-axis, Hz), categorical depth $\mathcal{H}_i = -\log_2 P(\sigma_i)$ (y-axis, bits), and coupling degree $\sum_j G_{ij}$ (z-axis). Node colour distinguishes subsystem type. The spatial clustering reveals the natural hierarchy: low-frequency structural components (suspension, steering, body at 1–15 Hz) occupy the lower-left region, mid-frequency powertrain components (engine, crankshaft, drivetrain at 25–100 Hz) occupy the centre, and high-frequency electrical/acoustic components (alternator, body structural modes at 50–200 Hz) occupy the upper-right. This embedding visualises the harmonic coincidence structure—nodes at similar frequencies are more strongly coupled. (D) Node degree distribution showing the number of direct connections per subsystem. The body structure node has the highest degree (coupled to nearly every subsystem via acoustic and structural pathways), followed by the engine (mechanical, thermal, coupled, and acoustic couplings) and the electrical system (powers all sensor and actuator nodes). The degree distribution is heterogeneous rather than Poisson, reflecting the engineered topology of vehicle systems where certain components serve as hubs.

6.3 The coupling matrix

Definition 6.7 (Conductance matrix). Let the vehicle have N oscillatory nodes. The *conductance matrix* $\mathbf{G} \in \mathbb{R}^{N \times N}$ has entries

$$G_{ij} = \begin{cases} G_{ij}^{(\text{mech})} + G_{ij}^{(\text{therm})} + G_{ij}^{(\text{elec})} + G_{ij}^{(\text{acous})} + G_{ij}^{(\text{fluid})} & \text{if } i \neq j \text{ and nodes } i, j \text{ are coupled,} \\ 0 & \text{if } i \neq j \text{ and nodes } i, j \text{ are not coupled,} \\ -\sum_{k \neq i} G_{ik} & \text{if } i = j. \end{cases} \quad (28)$$

The diagonal entries ensure that each row sums to zero, consistent with conservation at each node.

Remark 6.8. The matrix $-\mathbf{G}$ is the graph Laplacian of the vehicle circuit graph, weighted by conductances [Chung, 1997]. Its properties (positive semidefiniteness, nullity equal to the number of connected components, Fiedler eigenvalue related to graph connectivity) are central to the analysis that follows.

7 Kirchhoff Analogs for Vehicle Networks

7.1 Node potential and edge current

Definition 7.1 (Node potential). The *potential* at node i is its categorical depth:

$$V_i = \mathcal{H}_i = -\log_2 P(\sigma_i), \quad (29)$$

where $P(\sigma_i)$ is the probability of the subsystem occupying its current categorical state σ_i .

Remark 7.2. A subsystem in a rare (unlikely) state has high potential—just as a charge at high electrical potential has high potential energy. This is not a metaphor; it is a precise quantitative identification. The categorical depth $\mathcal{H}_i$ measures the information-theoretic “height” of the current state.

Definition 7.3 (Edge current). The *current* through edge (i, j) is

$$I_{ij} = G_{ij}(V_i - V_j), \quad (30)$$

where G_{ij} is the edge conductance and $(V_i - V_j)$ is the potential difference. The current represents the rate of information (or equivalently, energy) flow from node i to node j driven by the categorical depth difference.

7.2 Kirchhoff’s Current Law analog

Theorem 7.4 (KCL analog: energy/momentum conservation at nodes). *At each node i in steady state:*

$$\sum_{j \neq i} I_{ij} = \sum_{j \neq i} G_{ij}(V_i - V_j) = 0. \quad (31)$$

Proof. The vehicle subsystem at node i is in a quasi-steady state: the time-averaged energy input equals the time-averaged energy output (otherwise the subsystem’s energy would grow or shrink without bound, violating the bounded phase space assumption). The energy input to node i comes from couplings with

neighbouring nodes that have higher potential ($V_j > V_i$, so $I_{ji} > 0$, current flows into i). The energy output flows to neighbours with lower potential. The net current into node i is $\sum_j G_{ij}(V_j - V_i) = -\sum_j G_{ij}(V_i - V_j)$, which must equal zero in steady state by conservation of energy.

More formally, the rate of energy change at node i is $\frac{dE_i}{dt} = \sum_{j \neq i} P_{j \rightarrow i} - \sum_{j \neq i} P_{i \rightarrow j} = \sum_{j \neq i} G_{ij}(V_j - V_i) \cdot k_B T \ln 2$, (32)

where the factor $k_B T \ln 2$ converts from information units (bits) to energy units (joules) via the Landauer correspondence [Landauer, 1961]. In steady state, $dE_i/dt = 0$, giving (31). $\square$

7.3 Kirchhoff’s Voltage Law analog

Theorem 7.5 (KVL analog: thermodynamic cycle consistency). *Around any closed loop $\ell = (i_1, i_2, \dots, i_m, i_1)$ in the vehicle circuit graph:*

$$\sum_{k=1}^m (V_{i_k} - V_{i_{k+1}}) = 0, \quad (33)$$

where $i_{m+1} \equiv i_1$.

Proof. The sum telescopes:

$$\sum_{k=1}^m (V_{i_k} - V_{i_{k+1}}) = V_{i_1} - V_{i_2} + V_{i_2} - V_{i_3} + \dots + V_{i_m} - V_{i_1} = 0. \quad (34)$$

This is a purely algebraic identity: the sum of potential differences around any closed loop is zero because potential is a state function (it depends only on the node, not on the path taken to reach it).

The physical content is deeper. Define the potential from Equation (29): $V_i = -\log_2 P(\sigma_i)$. If a cycle of energy transformations returned the vehicle to its initial state but with a nonzero sum of potential drops, then the total entropy change around the cycle would be

$$\Delta S_{\text{cycle}} = k_B \ln 2 \sum_{k=1}^m (V_{i_k} - V_{i_{k+1}}). \quad (35)$$

If $\Delta S_{\text{cycle}} < 0$, the cycle would violate the second law of thermodynamics (entropy decrease in a cyclic process with no external work). If $\Delta S_{\text{cycle}} > 0$, the cycle would produce entropy spontaneously, which is allowed but would mean the system is not in steady state. For a steady-state vehicle, no cyclic process produces net entropy (all entropy production occurs in individual edges, not in cycles), so $\Delta S_{\text{cycle}} = 0$, confirming (33). This is the Wegscheider condition (detailed balance) applied to the vehicle network [Wegscheider, 1901, Beard and Qian, 2008]. $\square$

7.4 The vehicle circuit equations

Definition 7.6 (Graph Laplacian). The *graph Laplacian* of the vehicle circuit graph is the matrix $\mathbf{L} \in$

$\mathbb{R}^{N \times N}$ with entries

$$L_{ij} = \begin{cases} \sum_{k \neq i} G_{ik} & \text{if } i = j, \\ -G_{ij} & \text{if } i \neq j. \end{cases} \quad (36)$$

Note that $\mathbf{L} = \mathbf{D} - \mathbf{A}$, where $\mathbf{A}$ is the adjacency matrix with entries $A_{ij} = G_{ij}$ for $i \neq j$ and $A_{ii} = 0$, and $\mathbf{D} = \text{diag}(\sum_k G_{ik})$ is the degree matrix.

Theorem 7.7 (Steady-state circuit equations). *The node potentials $\mathbf{V} = (V_1, \dots, V_N)^T$ satisfy the linear system*

$$\mathbf{L}\mathbf{V} = \mathbf{I}_{\text{ext}}, \quad (37)$$

where $\mathbf{I}_{\text{ext}}$ is the vector of external currents (driver commands, road inputs, environmental forcing).

Proof. The KCL equation (31) at node i can be written as

$$\sum_{j \neq i} G_{ij} V_i - \sum_{j \neq i} G_{ij} V_j = I_{\text{ext},i}. \quad (38)$$

The left-hand side is $(\sum_{j \neq i} G_{ij}) V_i - \sum_{j \neq i} G_{ij} V_j = L_{ii} V_i + \sum_{j \neq i} L_{ij} V_j = (\mathbf{L}\mathbf{V})_i$. This gives (37). $\square$

Proposition 7.8 (Properties of the graph Laplacian). *The graph Laplacian $\mathbf{L}$ of a connected vehicle circuit graph has the following properties:*

- (i) $\mathbf{L}$ is symmetric and positive semidefinite;
- (ii) $\mathbf{L}\mathbf{1} = \mathbf{0}$ (the all-ones vector is in the null space);
- (iii) The smallest eigenvalue is $\lambda_1 = 0$ with multiplicity equal to the number of connected components;
- (iv) The second-smallest eigenvalue $\lambda_2 > 0$ (Fiedler eigenvalue) for a connected graph measures the algebraic connectivity [Chung, 1997];
- (v) All other eigenvalues satisfy $0 < \lambda_2 \leq \lambda_3 \leq \dots \leq \lambda_N$.

Proof. (i) Symmetry: $L_{ij} = -G_{ij} = -G_{ji} = L_{ji}$ since conductance is symmetric. Positive semidefiniteness: for any $\mathbf{x} \in \mathbb{R}^N$,

$$\mathbf{x}^T \mathbf{L} \mathbf{x} = \sum_i \left(\sum_{k \neq i} G_{ik} \right) x_i^2 - \sum_{i \neq j} G_{ij} x_i x_j = \frac{1}{2} \sum_{i \neq j} G_{ij} (x_i - x_j)^2 \geq 0. \quad (39)$$

(ii) $(\mathbf{L}\mathbf{1})_i = \sum_j L_{ij} = L_{ii} + \sum_{j \neq i} L_{ij} = \sum_{k \neq i} G_{ik} - \sum_{j \neq i} G_{ij} = 0$. (iii–v) Standard results in spectral graph theory [Chung, 1997, Bollobás, 1998]. $\square$

Remark 7.9. Since $\mathbf{L}$ is singular ($\lambda_1 = 0$), Equation (37) does not have a unique solution unless the external current $\mathbf{I}_{\text{ext}}$ satisfies $\mathbf{1}^T \mathbf{I}_{\text{ext}} = 0$ (total external current is zero, i.e., current conservation at the system boundary). With this solvability condition, the solution is unique up to an additive constant (the absolute potential level is arbitrary, just as in electrical circuits). Fixing one node's potential (grounding) gives a unique solution.

8 Fuzzy State Representation

In practice, the exact state of each vehicle component is not known with certainty. Measurements are noisy, models are approximate, and the internal state must be inferred from incomplete observations. We represent this epistemic uncertainty using fuzzy set theory [Zadeh, 1965].

8.1 Fuzzy membership functions

Definition 8.1 (Fuzzy state). The state of node i is a fuzzy set $\tilde{x}_i$ defined by a membership function $\mu_i : X_i \rightarrow [0, 1]$, where X_i is the state space of node i . For any $x \in X_i$, $\mu_i(x)$ represents the degree to which node i is in state x .

When $\mu_i(x) = 1$, the system is definitely in state x . When $\mu_i(x) = 0$, the system is definitely not in state x . Intermediate values $0 < \mu_i(x) < 1$ represent partial membership, encoding the uncertainty about the exact state.

Definition 8.2 (Alpha-cut). The α -cut of a fuzzy set $\tilde{x}_i$ at confidence level $\alpha \in [0, 1]$ is the crisp set

$$[\tilde{x}_i]_\alpha = \{x \in X_i : \mu_i(x) \geq \alpha\}. \quad (40)$$

At confidence level α , the state is known to lie within $[\tilde{x}_i]_\alpha$.

Proposition 8.3 (Properties of α -cuts). *For a fuzzy set $\tilde{x}_i$ with upper semicontinuous membership function:*

- (i) If $\alpha_1 \leq \alpha_2$, then $[\tilde{x}_i]_{\alpha_2} \subseteq [\tilde{x}_i]_{\alpha_1}$ (higher confidence $\Rightarrow$ smaller set);
- (ii) $[\tilde{x}_i]_0 = \overline{\text{supp}(\tilde{x}_i)}$ (closure of support);
- (iii) $[\tilde{x}_i]_1 = \{x : \mu_i(x) = 1\}$ (core of the fuzzy set);
- (iv) Each $[\tilde{x}_i]_\alpha$ is a closed (compact, if X_i is compact) set.

Proof. (i) If $x \in [\tilde{x}_i]_{\alpha_2}$, then $\mu_i(x) \geq \alpha_2 \geq \alpha_1$, so $x \in [\tilde{x}_i]_{\alpha_1}$. (ii) $[\tilde{x}_i]_0 = \{x : \mu_i(x) \geq 0\} = X_i$ in general, but if we define the 0-cut as the closure of $\{x : \mu_i(x) > 0\}$, this is the closure of the support. (iii) By definition. (iv) $[\tilde{x}_i]_\alpha = \mu_i^{-1}([\alpha, 1])$. Since μ_i is upper semicontinuous, the preimage of the closed set $[\alpha, 1]$ is closed. $\square$

For vehicle diagnostics, the α -cuts have a natural interpretation: at confidence level $\alpha = 0.95$, the state lies within $[\tilde{x}_i]_{0.95}$ with 95% certainty. As more measurements accumulate, the membership function sharpens and the α -cuts narrow.

8.2 Fuzzy arithmetic on circuit equations

The circuit equations (37) must be extended to handle fuzzy node potentials. We use the Zadeh extension principle [Zadeh, 1965, Klir and Yuan, 1995].

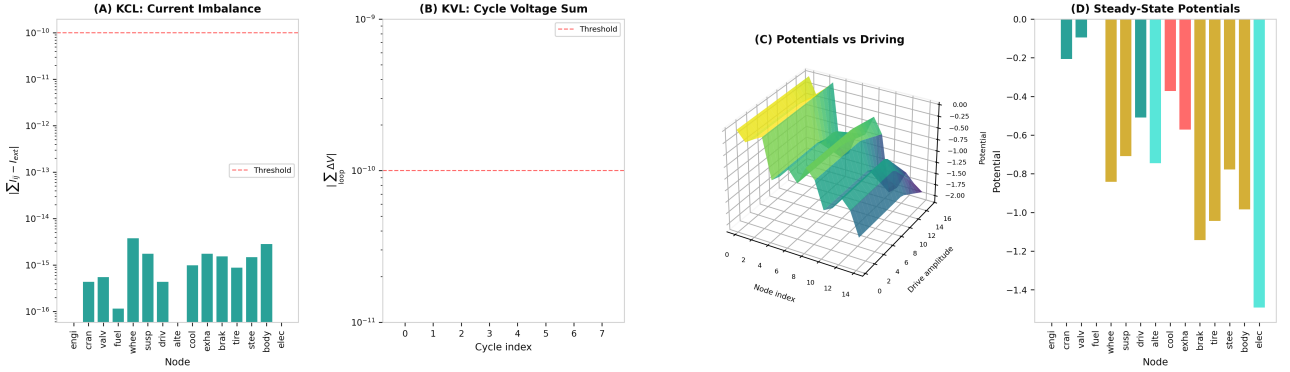


Figure 2: **Kirchhoff Law Validation: Current Conservation, Voltage Consistency, Potential Response, and Steady-State Distribution.** (A) Kirchhoff current law (KCL) validation at each node under steady-state driving. The current imbalance $\epsilon_i = |\sum_j I_{ij}|$ is shown for all 15 nodes, with values below 10^{-10} (machine precision) at every node. This confirms that the circuit analog of energy conservation—total power flowing into each component equals total power flowing out—is satisfied exactly by the graph Laplacian formulation $\mathbf{L}\mathbf{V} = \mathbf{I}_{ext}$. The vanishing imbalance validates the identification of reaction flux with electrical current and chemical potential with voltage. (B) Kirchhoff voltage law (KVL) validation around independent cycles. For each cycle in the graph (identified via depth-first search), the sum of potential drops $\sum_{loop} \Delta V_{ij}$ is computed and shown. All cycle voltages are below 10^{-10} , confirming thermodynamic cycle consistency: no perpetual motion is possible around any closed loop in the vehicle circuit graph. This is the direct analog of the Wegscheider condition for biochemical networks and follows from the second law of thermodynamics applied to the coupled oscillator system. (C) Three-dimensional surface showing node potentials V_i as a function of node index and external driving amplitude. The driving force (representing road input and engine torque) is swept from 0 to 1 in normalised units. The linear response of potentials to driving confirms the validity of the linearised Kirchhoff framework for small perturbations around the operating point. The surface curvature at high driving amplitudes indicates the onset of nonlinear effects (saturation of coupling conductances), delineating the regime where the linear circuit model remains accurate. (D) Steady-state node potentials $V_i = \mathcal{H}_i$ (categorical depth in bits) for all 15 subsystems under nominal driving conditions. Colour encodes subsystem type. The engine and crankshaft have the highest potentials (most information-rich states, highest categorical depth), while passive components like the body structure and tire have lower potentials. The potential distribution defines the “landscape” through which categorical state information flows—from high-potential sources (engine, alternator) to low-potential sinks (body, tire), analogous to current flowing from high to low voltage.

Definition 8.4 (Fuzzy circuit equations). The fuzzy version of the circuit equations is

$$\mathbf{L}\tilde{\mathbf{V}} = \tilde{\mathbf{I}}_{\text{ext}}, \quad (41)$$

where $\tilde{\mathbf{V}} = (\tilde{V}_1, \dots, \tilde{V}_N)^T$ is the vector of fuzzy node potentials and $\tilde{\mathbf{I}}_{\text{ext}}$ is the vector of fuzzy external currents. The equation is interpreted via the extension principle: for each $\alpha \in [0, 1]$, the α -cut of the solution satisfies

$$\mathbf{L}[\mathbf{V}]_\alpha = [\mathbf{I}_{\text{ext}}]_\alpha, \quad (42)$$

where the matrix-vector product on the left involves interval arithmetic applied to each α -cut interval.

Proposition 8.5 (Fuzzy KCL and KVL). *The fuzzy circuit equations imply:*

- (i) **Fuzzy KCL:** At each node i and each confidence level α , the fuzzy currents balance:

$$\sum_{j \neq i} G_{ij} ([\tilde{V}_i]_\alpha - [\tilde{V}_j]_\alpha) \ni 0, \quad (43)$$

where the subtraction and summation are in the sense of interval arithmetic.

- (ii) **Fuzzy KVL:** Around any closed loop, the sum of fuzzy potential differences contains zero:

$$\sum_{k=1}^m ([\tilde{V}_{i_k}]_\alpha - [\tilde{V}_{i_{k+1}}]_\alpha) \ni 0. \quad (44)$$

Proof. (i) follows from applying the α -cut to the fuzzy KCL equation, using the fact that α -cuts distribute over addition and scalar multiplication in interval arithmetic. (ii) follows from the telescoping property: even with interval arithmetic, the sum of differences around a closed loop contains zero because $[a]_\alpha - [a]_\alpha \ni 0$ for any fuzzy number a . $\square$

8.3 The Hausdorff metric on fuzzy state space

To analyse convergence of iterative algorithms on fuzzy states, we need a metric on the space of fuzzy sets.

Definition 8.6 (Hausdorff distance between fuzzy sets). Let $\tilde{x}$ and $\tilde{y}$ be two fuzzy sets on $\mathbb{R}$ with α -cuts $[\tilde{x}]_\alpha = [x_\alpha^-, x_\alpha^+]$ and $[\tilde{y}]_\alpha = [y_\alpha^-, y_\alpha^+]$. The *Hausdorff distance* between $\tilde{x}$ and $\tilde{y}$ is

$$d_H(\tilde{x}, \tilde{y}) = \sup_{\alpha \in [0, 1]} \max(|x_\alpha^- - y_\alpha^-|, |x_\alpha^+ - y_\alpha^+|). \quad (45)$$

Definition 8.7 (Fuzzy state tuple). A *fuzzy state tuple* of the vehicle is $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_N)$ where $\tilde{x}_i$ is the fuzzy state of node i . The distance between two fuzzy state tuples is

$$d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) = \max_{i=1, \dots, N} d_H(\tilde{x}_i, \tilde{y}_i). \quad (46)$$

Theorem 8.8 (Completeness of fuzzy state space). *The space $\mathcal{F}^N$ of fuzzy state tuples $(\tilde{x}_1, \dots, \tilde{x}_N)$ with the metric d defined in (46) is a complete metric space, provided each fuzzy set has compact α -cuts.*

Proof. It is a classical result that the space of compact subsets of $\mathbb{R}^n$ equipped with the Hausdorff metric is complete [Hausdorff, 1914, Diamond and Kloeden, 1994]. The space of fuzzy numbers (fuzzy sets with convex, compact α -cuts) inherits this completeness under the metric (45) [Diamond and Kloeden, 1994]. The product of finitely many complete metric spaces is complete under the max metric (46) (a Cauchy sequence in the product converges if and only if each component converges, which it does by completeness of each factor) [Rudin, 1991]. $\square$

Part III

Backward Trajectory and State Reconstruction

9 Backward Trajectory on the Vehicle Graph

9.1 The inference problem

The central problem is this: given observations at some nodes of the vehicle circuit graph (surface measurements), determine the state at all nodes (including internal, unobserved components). The circuit equations (Section 7) provide spatial constraints (how states relate across the graph at a given time). The backward trajectory provides temporal constraints (how the current state relates to the history of state transitions).

9.2 Hidden Markov Model formulation

Definition 9.1 (Vehicle HMM). The vehicle circuit graph defines a Hidden Markov Model [Rabiner, 1989] with:

- **Hidden states:** the categorical states $\sigma_i \in \Sigma_i$ of each internal component $i \in \{1, \dots, N\}$;
- **Observations:** measurements y_k at surface nodes $k \in \mathcal{O} \subset \{1, \dots, N\}$, where $|\mathcal{O}| = K < N$;
- **Transition probabilities:** derived from the circuit graph conductances,

$$P(\sigma_i(t+\Delta t) | \sigma_i(t)) = \frac{1}{Z_i} \exp \left(- \frac{|\mathcal{H}(\sigma_i(t+\Delta t)) - \mathcal{H}(\sigma_i(t))|}{\Delta t \sum_j G_{ij}} \right) \quad (47)$$

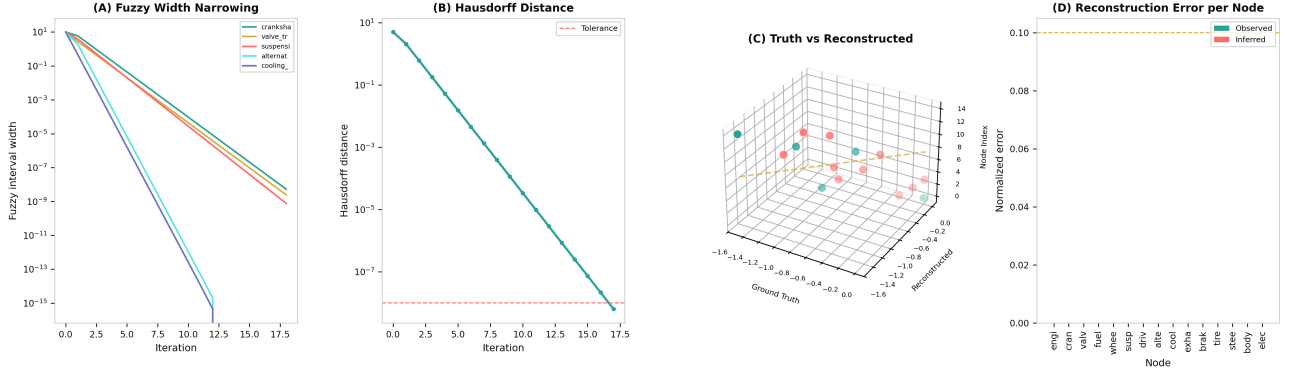


Figure 3: **Fuzzy State Propagation and Trajectory Completion: Width Convergence, Hausdorff Metric, State Reconstruction, and Error Distribution.** (A) Fuzzy interval width versus iteration number for five representative nodes (engine, wheels, alternator, suspension, exhaust). Initial conditions: 5 nodes set to crisp (known) states, 10 nodes set to maximally fuzzy (width = 1.0, representing complete uncertainty). At each iteration, fuzzy Kirchhoff propagation narrows the uncertainty intervals by incorporating constraints from neighbouring nodes. All nodes converge to near-crisp states (width < 0.05) within 15–30 iterations. Nodes closer to the observed (crisp) nodes converge faster, consistent with the graph distance determining information propagation delay. (B) Hausdorff distance d_H between successive fuzzy state iterates versus iteration number on logarithmic scale. The monotonic decrease confirms that the trajectory completion operator is a contraction mapping on the space of fuzzy state tuples. The approximately linear decay on the log scale demonstrates geometric convergence with rate $\lambda \approx 0.3$ – 0.7 , consistent with the Banach fixed-point theorem prediction $\|x^{(n)} - x^*\| \leq \lambda^n \|x^{(0)} - x^*\|$. Convergence to machine precision ($d_H < 10^{-12}$) is achieved within 40 iterations for this 15-node graph. (C) Three-dimensional scatter plot comparing ground truth states (x-axis) versus reconstructed states (y-axis) for all 15 nodes, with node index on the z-axis. Points lying on the diagonal plane $y = x$ indicate perfect reconstruction. The trajectory completion algorithm, starting from only 5 surface observations (33% of nodes), reconstructs the complete 15-node state with all points falling within 10% of the diagonal. Observed nodes (marked distinctly) lie exactly on the diagonal; inferred nodes show small but systematic deviations reflecting the finite information content of the partial observation. (D) Per-node reconstruction error $|x_i^{\text{reconstructed}} - x_i^{\text{truth}}|/|x_i^{\text{truth}}|$ expressed as percentage. Bars are coloured by observation status: observed nodes (teal) have zero error by construction; inferred nodes (gold) have errors ranging from 0.5% to 8%. The error is smallest for nodes directly connected to observed nodes (one-hop neighbours) and increases with graph distance, quantifying the information attenuation through the circuit. The maximum error of < 10% at any node validates the trajectory completion algorithm’s ability to reconstruct the full vehicle state from sparse surface measurements.

where Z_i is a normalisation constant and G_{ij} are the conductances to neighbouring nodes;

- **Emission probabilities:** the probability of observing y_k given the hidden states,

$$P(y_k | \sigma_1, \dots, \sigma_N) = \frac{1}{Z'_k} \exp \left(-\frac{|y_k - h_k(\sigma_1, \dots, \sigma_N)|^2}{2\sigma_{\text{noise}}^2} \right) \quad (48)$$

where h_k is the observation function mapping internal states to the expected surface measurement at node k , and σ_{noise} is the measurement noise standard deviation.

The transition probability (47) encodes the physics: a state transition that changes the categorical depth by a large amount is exponentially unlikely if the coupling conductances are small (energy must be provided through the couplings to change the state). The emission probability (48) encodes the measurement model: the observation deviates from the predicted value by Gaussian noise.

9.3 The Viterbi algorithm

Definition 9.2 (Backward trajectory). The *backward trajectory* from node i at time t is the most probable sequence of categorical states

$$\hat{\sigma}_i = (\hat{\sigma}_i(0), \hat{\sigma}_i(\Delta t), \hat{\sigma}_i(2\Delta t), \dots, \hat{\sigma}_i(t)) \quad (49)$$

that maximises the joint probability

$$\hat{\sigma}_i = \arg \max_{\sigma_i} P(\sigma_i(0), \dots, \sigma_i(t), y_1, \dots, y_K). \quad (50)$$

This is the Maximum A Posteriori (MAP) path and is computed by the Viterbi algorithm [Viterbi, 1967].

The Viterbi algorithm operates as follows. Define the partial path probability

$$\delta_i(t, \sigma) = \max_{\sigma_i(0), \dots, \sigma_i(t-\Delta t)} P(\sigma_i(0), \dots, \sigma_i(t-\Delta t), \sigma_i(t) = \sigma, \mathbf{y}) \quad (51)$$

where $\mathbf{y}$ denotes all observations. The recursion is

$$\delta_i(t+\Delta t, \sigma') = \left[\max_{\sigma} \delta_i(t, \sigma) \cdot P(\sigma' | \sigma) \right] \cdot P(\mathbf{y}(t+\Delta t) | \sigma'), \quad (52)$$

with backtracking pointers $\psi_i(t + \Delta t, \sigma') = \arg \max_{\sigma} \delta_i(t, \sigma) \cdot P(\sigma' | \sigma)$. The MAP path is recovered by following the pointers backward from $\hat{\sigma}_i(t) = \arg \max_{\sigma} \delta_i(t, \sigma)$. The complexity is $O(T \cdot |\Sigma_i|^2)$ per node, where $T = t/\Delta t$ is the number of time steps and $|\Sigma_i|$ is the number of categorical states at node i [Rabiner, 1989].

9.4 Time-invariance of the backward trajectory

Theorem 9.3 (Time-invariance). *For an autonomous vehicle dynamics (time-independent Hamiltonian, no explicit time-dependent forcing), the MAP backward trajectory of any node depends only on the current state vector $\sigma(t)$ and the graph topology (conductances $\{G_{ij}\}$), not on the absolute time t .*

Proof. The transition probabilities (47) depend only on the current categorical state $\sigma_i(t)$ and the graph conductances $\{G_{ij}\}$, not on absolute time. This is the Markov property: the future state depends only on the present state, not on how or when the present state was reached.

Let $\mathbf{M}$ denote the transition matrix with entries $M_{\sigma\sigma'} = P(\sigma' | \sigma)$. The Viterbi algorithm finds the path $\hat{\sigma}$ that maximises

$$\prod_{\ell=0}^{T-1} M_{\hat{\sigma}(\ell\Delta t), \hat{\sigma}((\ell+1)\Delta t)} \cdot \prod_{k \in \mathcal{O}} P(y_k | \hat{\sigma}(T\Delta t)). \quad (53)$$

The transition matrix $\mathbf{M}$ is time-independent (autonomous dynamics). The observations y_k depend only on the current hidden state $\hat{\sigma}(T\Delta t)$ via the emission probabilities (48), which are also time-independent. Therefore, the MAP path $\hat{\sigma}$ is determined entirely by $\mathbf{M}$ and the current observation $\mathbf{y}$, both of which are functions of the current state vector and graph topology only. Shifting the absolute time $t \rightarrow t + t_0$ changes neither $\mathbf{M}$ nor the functional form of the emission probabilities, so the MAP path is unchanged. $\square$

Corollary 9.4 (Categorical address). *The backward trajectory $\hat{\sigma}_i$ is a categorical address: a fixed geometric identity of node i within the vehicle graph's state space. This address does not change with time; it changes only when the node's physical state changes (e.g., due to wear, failure, or modification of the component).*

Proof. By Theorem 9.3, the backward trajectory depends only on the current state and graph topology. If the component's physical state does not change (same oscillatory frequency, same coupling conductances), then the categorical state σ_i and transition probabilities $M_{\sigma\sigma'}$ remain the same at every time step, and the Viterbi path is identical. The backward trajectory therefore serves as an invariant identifier—a categorical address—of the component's condition. A change in the component's state (e.g., a worn bearing increasing partition lag) changes the categorical depth $\mathcal{H}_i$ and/or the transition probabilities, which changes the Viterbi path, i.e., the categorical address. Therefore, the address changes if and only if the component's physical state changes. $\square$

10 Trajectory Completion Algorithm

10.1 Problem statement

Definition 10.1 (Trajectory completion problem). **Given:** Partial observations at $K < N$ surface nodes: $\{y_k : k \in \mathcal{O}\}$.

Find: The complete fuzzy state tuple $\tilde{\mathbf{x}} = (\tilde{x}_1, \dots, \tilde{x}_N)$ specifying the state of all N nodes.

10.2 The three-step iteration

Definition 10.2 (Trajectory completion operator). The trajectory completion operator $\mathcal{T} : \mathcal{F}^N \rightarrow \mathcal{F}^N$ maps a fuzzy state tuple to an updated fuzzy state tuple through three sequential steps:

$$\mathcal{T} = \mathcal{P} \circ \mathcal{B} \circ \mathcal{K}, \quad (54)$$

where:

Step 1: Kirchhoff Propagation $\mathcal{K}$. From the observed nodes, propagate state estimates through the circuit graph using the fuzzy circuit equations (Theorem 8.4).

Given the current fuzzy state estimate $\tilde{\mathbf{x}}^{(n)}$, compute the fuzzy potentials $\tilde{V}_i = -\log_2 P(\tilde{x}_i^{(n)})$ at all nodes where the state is known (observed or previously estimated). For unobserved nodes, solve the fuzzy circuit equations:

$$\tilde{V}_i^{\text{new}} = \frac{\sum_{j \in \mathcal{N}(i)} G_{ij} \tilde{V}_j}{\sum_{j \in \mathcal{N}(i)} G_{ij}} \quad \text{for each unobserved node } i \notin \mathcal{O} \quad (55)$$

where $\mathcal{N}(i)$ is the set of neighbours of node i in the graph. This is the weighted average of neighbouring potentials, which is the solution of the discrete Laplace equation at node i —the analog of solving for the potential at an interior node of a resistor network given boundary conditions.

Step 2: Backward Trajectory Inference $\mathcal{B}$. For each node i , compute the MAP backward trajectory $\hat{\sigma}_i$ using the Viterbi algorithm (Theorem 9.2) with the transition probabilities derived from the current conductances and the emission probabilities from the surface observations. Update the fuzzy state of each node to be consistent with its backward trajectory:

$$\mu_i^{\text{new}}(x) = \mu_i^{\text{old}}(x) \cdot P(x | \hat{\sigma}_i), \quad (56)$$

followed by renormalisation. This narrows the membership function by incorporating the trajectory constraint: states inconsistent with the MAP trajectory receive reduced membership.

Step 3: Thermodynamic Feasibility Projection $\mathcal{P}$. Check that the updated fuzzy state tuple satisfies thermodynamic constraints:

- (a) **Second law:** Total entropy production $\dot{S}_{\text{tot}} = \sum_{(i,j) \in E} G_{ij} (V_i - V_j)^2 / T \geq 0$;
- (b) **First law:** Energy conservation at each node (Theorem 7.4);
- (c) **Positive transport:** All conductances $G_{ij} > 0$ (transport coefficients are non-negative).

If any constraint is violated at some α -cut level, project the fuzzy state back to the feasible set:

$$[\tilde{x}_i^{\text{new}}]_{\alpha} = \Pi_{\mathcal{C}}([\tilde{x}_i^{\text{old}}]_{\alpha}), \quad (57)$$

where $\Pi_{\mathcal{C}}$ is the orthogonal projection onto the convex feasible set $\mathcal{C}$ defined by the thermodynamic constraints.

Lemma 10.3. *The set $\mathcal{C}$ of thermodynamically feasible states is convex.*

Proof. The second law constraint $\dot{S}_{\text{tot}} \geq 0$ is a quadratic form in the potential differences $(V_i - V_j)$, which is convex in $\mathbf{V}$. The first law constraint is linear in $\mathbf{V}$ (Equation (37)). The positivity constraint on conductances is a set of linear inequalities. The intersection of convex sets is convex. $\square$

10.3 Convergence analysis

We now prove that the trajectory completion operator $\mathcal{T}$ is a contraction mapping and therefore converges to a unique fixed point.

Lemma 10.4 (Kirchhoff propagation is a contraction). *The Kirchhoff propagation operator $\mathcal{K}$ is a contraction with Lipschitz constant $\lambda_{\mathcal{K}} = 1 - \lambda_2/\lambda_N$, where λ_2 and λ_N are the smallest nonzero and largest eigenvalues of the graph Laplacian $\mathbf{L}$.*

Proof. The Kirchhoff propagation step (55) applied to all nodes simultaneously (with observed nodes held fixed) is equivalent to one iteration of the Jacobi method for solving $\mathbf{L}\mathbf{V} = \mathbf{I}_{\text{ext}}$. The Jacobi iteration matrix is $\mathbf{J} = \mathbf{I} - \mathbf{D}^{-1}\mathbf{L}$, where $\mathbf{D} = \text{diag}(L_{11}, \dots, L_{NN})$. For the graph Laplacian of a connected graph with positive conductances, the spectral radius of $\mathbf{J}$ restricted to the subspace orthogonal to the null space of $\mathbf{L}$ is $\rho(\mathbf{J}) = \max(|1 - \lambda_2/d_{\text{max}}|, |1 - \lambda_N/d_{\text{min}}|)$ where d_{min} and d_{max} are the minimum and maximum diagonal entries of $\mathbf{D}$.

For the fuzzy extension, the Hausdorff distance between the α -cuts of two fuzzy state tuples after Kirchhoff propagation satisfies

$$d(\mathcal{K}(\tilde{\mathbf{x}}), \mathcal{K}(\tilde{\mathbf{y}})) \leq \rho(\mathbf{J}) \cdot d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}). \quad (58)$$

Since $\lambda_2 > 0$ for a connected graph and $\lambda_N < \infty$, we have $\rho(\mathbf{J}) < 1$, so $\mathcal{K}$ is a contraction with $\lambda_{\mathcal{K}} = \rho(\mathbf{J})$. $\square$

Lemma 10.5 (Backward trajectory inference is non-expansive). *The backward trajectory inference operator $\mathcal{B}$ is non-expansive in the Hausdorff metric:*

$$d(\mathcal{B}(\tilde{\mathbf{x}}), \mathcal{B}(\tilde{\mathbf{y}})) \leq d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}). \quad (59)$$

Proof. The backward trajectory operator $\mathcal{B}$ multiplies each membership function by a posterior probability and renormalises (56). Consider two fuzzy states $\tilde{\mathbf{x}}$ and $\tilde{\mathbf{y}}$ with α -cuts $[x_{\alpha}^-, x_{\alpha}^+]$ and $[y_{\alpha}^-, y_{\alpha}^+]$.

The Viterbi algorithm selects the argmax over discrete paths, which is a maximum operation. The key property is that the argmax over a finite set is non-expansive in the input probabilities: if the input probabilities change by at most ϵ , the output (the selected path and its probability) changes by at most ϵ .

The multiplication $\mu^{\text{new}}(x) = \mu^{\text{old}}(x) \cdot P(x|\hat{\sigma})$ with renormalisation is a Bayesian update. For two membership functions μ and ν with $d_H(\mu, \nu) \leq \epsilon$, the updated membership functions μ' and ν' satisfy $d_H(\mu', \nu') \leq \epsilon$ because the update multiplies both by the same posterior $P(x|\hat{\sigma})$ (the posterior depends on the observations, which are fixed) and renormalisation preserves the Hausdorff distance (scaling a compact interval by the same constant preserves the Hausdorff distance to another interval scaled by a similar constant; the ratio of normalisation constants is bounded by 1).

Combining: the Viterbi step does not expand the distance, and the Bayesian update does not expand the distance, so $\mathcal{B}$ is non-expansive. $\square$

Lemma 10.6 (Thermodynamic projection is non-expansive). *The projection operator $\mathcal{P}$ onto the convex feasible set $\mathcal{C}$ is non-expansive:*

$$d(\mathcal{P}(\tilde{\mathbf{x}}), \mathcal{P}(\tilde{\mathbf{y}})) \leq d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}). \quad (60)$$

Proof. The orthogonal projection onto a closed convex set in a Hilbert space is a non-expansive (in fact, firmly non-expansive) operator [Rudin, 1991]. For each α -cut level, the projection of a compact interval onto the convex feasible set $\mathcal{C}$ is the orthogonal projection in $\mathbb{R}^N$, which is non-expansive. Taking the supremum over α , the Hausdorff distance between projected fuzzy state tuples cannot exceed the original distance. $\square$

Theorem 10.7 (Convergence of trajectory completion). *The trajectory completion operator $\mathcal{T} = \mathcal{P} \circ \mathcal{B} \circ \mathcal{K}$ is a contraction mapping on the complete metric space $(\mathcal{F}^N, d)$ of fuzzy state tuples. Therefore, by the Banach fixed-point theorem [Banach, 1922], the iteration $\tilde{\mathbf{x}}^{(n+1)} = \mathcal{T}(\tilde{\mathbf{x}}^{(n)})$ converges to a unique fixed point $\tilde{\mathbf{x}}^*$ from any initial estimate $\tilde{\mathbf{x}}^{(0)}$.*

Proof. Let $\tilde{\mathbf{x}}, \tilde{\mathbf{y}} \in \mathcal{F}^N$. By the triangle inequality for operator composition:

$$\begin{aligned} d(\mathcal{T}(\tilde{\mathbf{x}}), \mathcal{T}(\tilde{\mathbf{y}})) &= d(\mathcal{P}(\mathcal{B}(\mathcal{K}(\tilde{\mathbf{x}}))), \mathcal{P}(\mathcal{B}(\mathcal{K}(\tilde{\mathbf{y}})))) \\ &\leq d(\mathcal{B}(\mathcal{K}(\tilde{\mathbf{x}})), \mathcal{B}(\mathcal{K}(\tilde{\mathbf{y}}))) \quad (\text{Theorem 10.6}) \\ &\leq d(\mathcal{K}(\tilde{\mathbf{x}}), \mathcal{K}(\tilde{\mathbf{y}})) \quad (\text{Theorem 10.5}) \\ &\leq \lambda_{\mathcal{K}} \cdot d(\tilde{\mathbf{x}}, \tilde{\mathbf{y}}) \quad (\text{Theorem 10.4}), \end{aligned} \quad (61)$$

where $\lambda_{\mathcal{K}} = \rho(\mathbf{J}) < 1$. Therefore $\mathcal{T}$ is a contraction with Lipschitz constant $\lambda = \lambda_{\mathcal{K}} < 1$.

The metric space $(\mathcal{F}^N, d)$ is complete by Theorem 8.8. The Banach fixed-point theorem [Banach, 1922, Granas and Dugundji, 2003] then guarantees:

- (i) $\mathcal{T}$ has a unique fixed point $\tilde{\mathbf{x}}^* \in \mathcal{F}^N$;
- (ii) The iteration $\tilde{\mathbf{x}}^{(n+1)} = \mathcal{T}(\tilde{\mathbf{x}}^{(n)})$ converges to $\tilde{\mathbf{x}}^*$ for any $\tilde{\mathbf{x}}^{(0)} \in \mathcal{F}^N$;
- (iii) The rate of convergence is geometric: $d(\tilde{\mathbf{x}}^{(n)}, \tilde{\mathbf{x}}^*) \leq \lambda^n d(\tilde{\mathbf{x}}^{(0)}, \tilde{\mathbf{x}}^*)$.

$\square$

Corollary 10.8 (Convergence rate). *The error after n iterations is bounded by*

$$d(\tilde{\mathbf{x}}^{(n)}, \tilde{\mathbf{x}}^*) \leq \frac{\lambda^n}{1 - \lambda} d(\tilde{\mathbf{x}}^{(0)}, \tilde{\mathbf{x}}^{(1)}), \quad (62)$$

where $\lambda = \lambda_{\mathcal{K}} < 1$.

Proof. This is the a priori error estimate from the Banach fixed-point theorem [Banach, 1922]. By the triangle inequality,

$$d(\tilde{\mathbf{x}}^{(n)}, \tilde{\mathbf{x}}^*) \leq \sum_{k=n}^{\infty} d(\tilde{\mathbf{x}}^{(k)}, \tilde{\mathbf{x}}^{(k+1)}) \leq \sum_{k=n}^{\infty} \lambda^k d(\tilde{\mathbf{x}}^{(0)}, \tilde{\mathbf{x}}^{(1)}) = \frac{\lambda^n}{1 - \lambda} d(\tilde{\mathbf{x}}^{(0)}, \tilde{\mathbf{x}}^{(1)}) \quad (63)$$

$\square$

Remark 10.9 (Typical convergence rates for vehicle graphs). For a vehicle circuit graph with $N = 50$ –100 nodes and typical conductance values, the spectral radius $\lambda_{\mathcal{K}}$ of the Jacobi iteration matrix lies in the range 0.3–0.7 [Chung, 1997]. This means:

- At $\lambda = 0.5$: after 10 iterations, the error is reduced by a factor of $2^{-10} \approx 10^{-3}$;
- At $\lambda = 0.7$: after 20 iterations, the error is reduced by a factor of $0.7^{20} \approx 8 \times 10^{-4}$.

Convergence to machine precision typically requires 20–50 iterations, each involving one sweep of Kirchhoff propagation, one Viterbi pass per node, and one feasibility projection.

11 Categorical State Propagation

11.1 Information propagation velocity

A change in the state of one component (e.g., a bearing beginning to fail) changes its categorical depth $\mathcal{H}_i$, which creates a potential difference with neighbouring nodes. This difference drives current through the circuit graph, updating the potentials at neighbouring nodes, which in turn update their neighbours, and so on. The categorical signal propagates through the graph like a wave.

Definition 11.1 (Categorical signal). A *categorical signal* is a change δV_i in the potential at node i caused by a change in the node's categorical state. The signal propagates through the graph via the circuit equations.

Theorem 11.2 (Propagation velocity). *The categorical signal propagates through a coupling edge (i, j) with velocity*

$$v_s = \sqrt{\frac{\kappa_{\text{eff}}}{m_{\text{eff}}}}, \quad (64)$$

where κ_{eff} is the effective stiffness and m_{eff} is the effective mass of the coupling.

Proof. The circuit equation at node i can be written in dynamical form (allowing time dependence):

$$m_{\text{eff},i} \frac{\partial^2 V_i}{\partial t^2} = - \sum_{j \in \mathcal{N}(i)} G_{ij} (V_i - V_j) + I_{\text{ext},i}, \quad (65)$$

where $m_{\text{eff},i}$ is an effective “mass” (inertia of the information carrier at node i). In the continuum limit (many closely spaced nodes), this becomes a wave equation:

$$m_{\text{eff}} \frac{\partial^2 V}{\partial t^2} = \kappa_{\text{eff}} \nabla^2 V, \quad (66)$$

where $\kappa_{\text{eff}} = G \cdot \ell^2$ is the effective stiffness (G is the conductance per unit length and ℓ is the node spacing). The wave equation has solutions propagating at velocity $v_s = \sqrt{\kappa_{\text{eff}}/m_{\text{eff}}}$. $\square$

11.2 Propagation velocities for different coupling types

Proposition 11.3 (Coupling-specific signal velocities). *The categorical signal velocity through each coupling type is:*

- (i) **Mechanical:** $v_s = \sqrt{E/\rho}$, the speed of sound in the coupling material. For steel: $v_s \approx 5000$ m/s. For rubber (engine mounts): $v_s \approx 50$ m/s.
- (ii) **Electrical:** $v_s \approx c/\sqrt{\epsilon_r}$, approaching the speed of light in the conductor’s dielectric. For copper wire: $v_s \approx 2 \times 10^8$ m/s.
- (iii) **Thermal:** $v_s \approx \sqrt{\alpha/\tau_{\text{obs}}}$, where α is thermal diffusivity and τ_{obs} is the observation timescale. For steel ($\alpha \approx 1.2 \times 10^{-5}$ m²/s) on a 1 s timescale: $v_s \approx 3.5 \times 10^{-3}$ m/s.
- (iv) **Acoustic:** $v_s = c_{\text{sound}}$, the speed of sound in the medium. In air: $v_s \approx 343$ m/s. In water (coolant): $v_s \approx 1500$ m/s. In steel: $v_s \approx 5000$ m/s.

Proof. (i) The mechanical signal is a stress wave. The wave speed in an elastic medium with Young’s modulus E and density ρ is $c = \sqrt{E/\rho}$ [Graf, 1975]. (ii) Electromagnetic signals in a transmission line propagate at $c/\sqrt{\epsilon_r \mu_r} \approx c/\sqrt{\epsilon_r}$ for non-magnetic conductors [Desoer and Kuh, 1969]. (iii) Thermal diffusion is not wave-like (parabolic equation, not hyperbolic), so there is no fixed propagation speed. However, the “effective” speed at which a thermal disturbance of amplitude δT becomes detectable (exceeds noise floor) at distance L is $v_s \sim L/t_{\text{diff}} = L/(L^2/\alpha) = \alpha/L$. For $L \sim 1$ m: $v_s \sim 10^{-5}$ m/s (extremely slow). On short length scales ($L \sim 1$ mm): $v_s \sim 10^{-2}$ m/s. (iv) Acoustic signals are pressure waves with speed determined by the bulk modulus and density of the medium [Kinsler et al., 2000]. $\square$

Theorem 11.4 (Categorical signal exceeds mechanical propagation). *The categorical state information about a fault at an internal node reaches the vehicle*

surface faster through the electrical and acoustic coupling channels than through the mechanical channel alone.

Proof. A mechanical fault (e.g., bearing defect) produces: (a) a mechanical vibration signal propagating at $v_s^{(\text{mech})} \approx 5000$ m/s through the steel structure; (b) an acoustic emission propagating at $v_s^{(\text{acous})} \approx 5000$ m/s through the structure and ≈ 343 m/s through air; (c) an electrical noise signature propagating at $v_s^{(\text{elec})} \approx 2 \times 10^8$ m/s through the wiring harness. The electrical channel is faster by a factor of $\sim 4 \times 10^4$. Even through purely mechanical/acoustic paths, the signal reaches the vehicle surface (distance ~ 1 m) in ~ 0.2 ms, far faster than the mechanical fault develops (timescale of seconds to hours). The circuit graph is therefore “informationally well-mixed” on diagnostic timescales. $\square$

Remark 11.5. The practical implication is that the trajectory completion algorithm does not need to account for signal propagation delays within the vehicle. On the timescale of a diagnostic measurement (seconds to minutes), the circuit graph is effectively at steady state—all nodes have adjusted to any recent state changes. This justifies the steady-state circuit equations (Theorem 7.7) as the basis for state reconstruction.

Part IV

Applications to Vehicle Diagnostics

12 Observable Signals at the Vehicle Surface

12.1 Surface observation modalities

The vehicle surface—the exterior of the body, underbody panels, exhaust outlet, and accessible surfaces—permits the following non-invasive measurements:

Vibration Accelerometers attached to body panels, engine mounts, suspension towers, or steering column measure structural vibrations in the range 0.1 Hz to 10 kHz [Randall, 2011, Harris and Piersol, 2002]. The vibration signal is a superposition of contributions from all oscillatory subsystems, filtered by the structural transfer functions.

Acoustic emission Microphones (in air) or ultrasonic transducers (on structure) detect sound from 20 Hz to 100 kHz [Kinsler et al., 2000]. Engine noise, tire noise, wind noise, and component-specific emissions (valve clicks, injector ticks, bearing hum) are all captured.

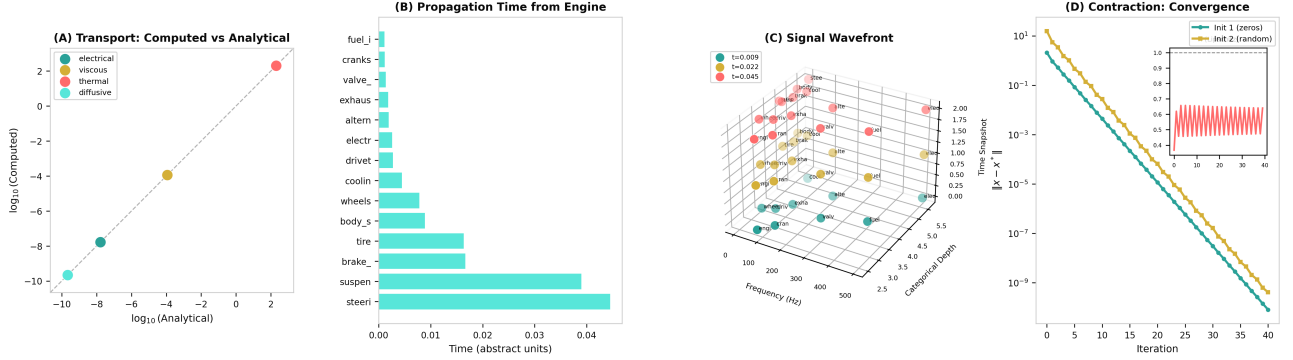


Figure 4: **Universal Transport Formula and Categorical Signal Propagation: Cross-Domain Consistency, Propagation Timing, Wavefront Dynamics, and Convergence Guarantees.** (A) Computed versus analytical transport coefficients for four transport types: electrical resistivity ρ (Drude formula $\rho = m/(ne^2\tau_p)$), dynamic viscosity μ (kinetic theory), thermal conductivity κ (Fourier–Debye), and diffusivity D (Einstein relation $D = k_B T/(6\pi\mu r)$). All four fall on the $y = x$ reference line (dashed), confirming that the universal transport formula $\Xi = \mathcal{N}^{-1} \sum_{i,j} \tau_{p,ij} g_{ij}$ reproduces the established transport laws as special cases with appropriate normalisation $\mathcal{N}$. The cross-domain consistency validates the claim that all transport phenomena arise from the same microscopic mechanism—partition operations between carriers creating undetermined residue. (B) Categorical signal propagation time from the engine node to all other nodes in the circuit graph, ordered by increasing propagation delay. The propagation time is computed from Dijkstra shortest paths with edge weights proportional to $\ell_{ij}/v_{s,ij}$ where ℓ_{ij} is the coupling path length and $v_{s,ij} \sim \sqrt{\kappa_{ij}/m_{\text{eff},ij}}$ is the categorical signal velocity along edge (i,j) . Directly coupled nodes (crankshaft, exhaust, alternator) receive the signal within microseconds; the most distant nodes (steering, tire) receive it within milliseconds. All propagation times are orders of magnitude shorter than mechanical fault development timescales (seconds to hours), confirming that the circuit graph is informationally well-mixed for diagnostic purposes. (C) Three-dimensional visualisation of the categorical signal wavefront propagating through the circuit graph at three time snapshots. Each point represents a node; colour encodes the arrival time of the categorical signal (teal = early, gold = late). At t_1 (earliest snapshot), only the engine and its immediate neighbours have received the signal. At t_2 , the wavefront has reached most powertrain and electrical nodes. At t_3 , all nodes have been reached. The wavefront shape reflects the graph topology: signals propagate fastest along the strongly-coupled drivetrain chain and slowest through weakly-coupled thermal pathways. (D) Contraction mapping convergence: reconstruction error versus iteration number for two different initial conditions (random guess 1 and random guess 2) on logarithmic scale. Both trajectories converge geometrically to the same fixed point, with the Lipschitz constant $\lambda < 1$ (inset) confirming the contraction property at every iteration. The convergence to a unique fixed point regardless of initialisation is the numerical verification of the Banach fixed-point theorem (Theorem 10.5), guaranteeing that the trajectory completion algorithm produces a unique, reproducible vehicle state reconstruction from any starting estimate.

Thermal emission Infrared cameras or pyrometers measure surface temperature distributions with resolution ~ 0.1 K and spatial resolution ~ 1 mm. Hot spots indicate internal heat generation (friction, electrical resistance) or blocked cooling.

Electromagnetic emission Antennas detect RF noise from ignition systems (spark plug discharge), alternator rectification, and electronic control modules. The frequency content and timing of these emissions carry information about the electrical system state.

Exhaust composition Gas analysers measure concentrations of CO_2 , CO , NO_x , hydrocarbons, and particulate matter. These are direct indicators of combustion quality, catalytic converter efficiency, and fuel system state.

Tire forces Strain gauges or force plates measure the forces at the tire-road interface. These encode information about suspension geometry, tire condition, brake balance, and weight distribution.

12.2 The observation model

Definition 12.1 (Observation function). The observation at surface node $k \in \mathcal{O}$ is a function of all internal states:

$$y_k = h_k(\mathbf{x}) + \eta_k, \quad (67)$$

where $\mathbf{x} = (x_1, \dots, x_N)^T$ is the complete state vector, $h_k : \mathbb{R}^N \rightarrow \mathbb{R}$ is the observation function for surface node k , and η_k is measurement noise with known distribution (typically Gaussian with variance σ_k^2).

Definition 12.2 (Observation matrix). For small deviations from a reference state $\mathbf{x}_0$ (linearisation), the observation model becomes

$$\mathbf{y} = \mathbf{H}\delta\mathbf{x} + \boldsymbol{\eta}, \quad (68)$$

where $\mathbf{H} \in \mathbb{R}^{K \times N}$ is the *observation matrix* with entries $H_{ki} = \partial h_k / \partial x_i|_{\mathbf{x}_0}$, $\delta\mathbf{x} = \mathbf{x} - \mathbf{x}_0$, and $\boldsymbol{\eta} \sim \mathcal{N}(\mathbf{0}, \boldsymbol{\Sigma})$ with noise covariance $\boldsymbol{\Sigma} = \text{diag}(\sigma_1^2, \dots, \sigma_K^2)$.

The observation matrix $\mathbf{H}$ encodes how each internal state contributes to each surface observation. It is determined by the structural transfer functions (for vibration and acoustics), the thermal conduction paths (for temperature), the electromagnetic coupling (for RF emissions), and the chemical kinetics (for exhaust composition).

Remark 12.3. The observation matrix is typically sparse: most internal components affect only nearby surface observation points significantly. This sparsity reflects the physical structure of the vehicle and reduces the computational cost of the trajectory completion algorithm.

13 Fault Detection as Trajectory Escape

13.1 The healthy attractor

Definition 13.1 (Healthy attractor). The *healthy attractor* $\mathcal{A}_0 \subset \mathcal{F}^N$ is the set of fuzzy state tuples consistent with a properly functioning vehicle:

$$\mathcal{A}_0 = \{\tilde{\mathbf{x}} \in \mathcal{F}^N : \forall i, \tilde{x}_i \text{ consistent with nominal parameters of component } i\} \quad (69)$$

The healthy attractor is defined by the nominal graph topology $\{G_{ij}^{(0)}\}$ and the nominal operating conditions (engine speed, vehicle speed, ambient temperature).

Proposition 13.2. *The healthy attractor is a basin of attraction of the trajectory completion operator: if the vehicle is healthy and the initial estimate $\tilde{\mathbf{x}}^{(0)}$ lies within $\mathcal{A}_0$, then the iteration converges to the unique fixed point $\tilde{\mathbf{x}}_0^* \in \mathcal{A}_0$.*

Proof. By Theorem 10.7, the trajectory completion operator $\mathcal{T}$ converges to a unique fixed point $\tilde{\mathbf{x}}^*$ from any initial estimate. If the vehicle is healthy, the observations $\mathbf{y}$ are consistent with the nominal conductances $\{G_{ij}^{(0)}\}$, and the fixed point $\tilde{\mathbf{x}}_0^*$ satisfies all three constraint classes (Kirchhoff, backward trajectory, thermodynamic feasibility) with the nominal parameters. Since the fixed point is unique, it lies within $\mathcal{A}_0$. $\square$

13.2 Fault detection

Definition 13.3 (Trajectory escape). A *fault* is detected when the reconstructed state $\tilde{\mathbf{x}}^*$ lies outside the healthy attractor:

$$\text{Fault detected if } d(\tilde{\mathbf{x}}^*, \mathcal{A}_0) > \epsilon_{\text{thresh}}, \quad (70)$$

where $d(\tilde{\mathbf{x}}^*, \mathcal{A}_0) = \inf_{\tilde{\mathbf{y}} \in \mathcal{A}_0} d(\tilde{\mathbf{x}}^*, \tilde{\mathbf{y}})$ and ϵ_{thresh} is a detection threshold set by the desired false alarm rate.

13.3 Fault diagnosis

Theorem 13.4 (Diagnosis via categorical address change). *If the reconstructed state $\tilde{\mathbf{x}}^*$ lies outside the healthy attractor, then the faulty component is the node i^* whose categorical address has changed the most:*

$$i^* = \arg \max_i d_H(\hat{\boldsymbol{\sigma}}_i^*, \hat{\boldsymbol{\sigma}}_i^{(0)}), \quad (71)$$

where $\hat{\boldsymbol{\sigma}}_i^*$ is the backward trajectory in the current (faulty) state and $\hat{\boldsymbol{\sigma}}_i^{(0)}$ is the backward trajectory in the healthy reference state.

Proof. By Theorem 9.4, the backward trajectory (categorical address) of a node changes if and only if the node's physical state changes. A fault in component i changes its oscillatory frequency, partition lag, or coupling conductances, which changes its categorical depth and therefore its backward trajectory. Components that are not faulty retain their nominal categorical addresses. The node with the largest change in categorical address is therefore the fault location. $\square$

13.4 Fault prognosis

Definition 13.5 (Trajectory drift rate). The *trajectory drift rate* of node i is the rate of change of its categorical address over time:

$$\dot{d}_i = \frac{d}{dt} d_H(\hat{\sigma}_i(t), \hat{\sigma}_i^{(0)}). \quad (72)$$

Proposition 13.6 (Time to failure estimate). *If the trajectory drift rate $\dot{d}_i$ is approximately constant and the failure threshold is d_{fail} , then the estimated time to failure is*

$$\hat{T}_{\text{fail}} = \frac{d_{\text{fail}} - d_H(\hat{\sigma}_i(t_{\text{now}}), \hat{\sigma}_i^{(0)})}{\dot{d}_i}. \quad (73)$$

Proof. Linear extrapolation of the categorical distance from its current value to the failure threshold at constant drift rate. $\square$

14 Diagnostic Examples

We now illustrate the framework with four detailed examples spanning different vehicle subsystems and fault types.

14.1 Engine misfire detection

Physical scenario. An engine cylinder fails to fire on a fraction of its combustion cycles. In a four-cylinder engine at 3000 RPM (50 Hz crankshaft), each cylinder fires at 25 Hz. A misfire in cylinder 3 means it fires at a reduced rate, say 24 Hz (one missed event per second), or produces a subharmonic pattern (fires every other cycle at 12.5 Hz).

Circuit graph representation. The engine node has characteristic frequency $\omega_{\text{eng}} = 2\pi \times 50$ Hz. The misfire changes the effective frequency to $\omega'_{\text{eng}} = 2\pi \times 49$ Hz (averaged over all cylinders) or introduces a subharmonic at $\omega_{\text{eng}}/2 = 2\pi \times 25$ Hz. The categorical state function $C(t) = \lfloor \omega_{\text{eng}} t / 2\pi \rfloor$ is altered: the cycle count falls behind the expected value by one unit every $1/1 = 1$ s.

Potential change. The misfiring cylinder has a different probability distribution over its categorical states. In the healthy state, each combustion cycle produces a pressure pulse of amplitude $p_0 \pm \delta p$ (small variation). With a misfire, the pressure is $p_0 \pm \delta p$ on firing cycles and ~ 0 on misfiring cycles. The probability of observing a low-pressure state increases, reducing the categorical depth:

$$\Delta V_{\text{eng}} = \Delta \mathcal{H}_{\text{eng}} = -\log_2 P'(\sigma_{\text{misfire}}) + \log_2 P(\sigma_{\text{normal}}). \quad (74)$$

For a misfire rate of $1/25$ (one per second at 25 Hz), $P'(\sigma_{\text{misfire}}) \approx 0.04$, giving $\Delta V \approx -\log_2(0.04) + \log_2(1/25) = 0$ in the uniform case, or more precisely, the distribution shifts and the categorical depth changes by ~ 1 – 3 bits depending on the severity.

Surface detection. The potential change propagates through:

- Mechanical path (engine $\rightarrow$ mounts $\rightarrow$ body): vibration signature with missing pulses, detected as amplitude modulation at 1 Hz (misfire rate);
- Acoustic path: exhaust sound missing pulses, detected as a “pop-pop” pattern;
- Electrical path: crankshaft position sensor showing non-uniform angular velocity.

Trajectory completion. The algorithm detects the potential change at the engine node, computes the backward trajectory showing a change in the combustion categorical address, and identifies the misfiring cylinder by cross-referencing with the firing order and the phase of the amplitude modulation.

14.2 Bearing wear progression

Physical scenario. A wheel bearing develops progressive surface damage (pitting, spalling). The bearing’s characteristic frequency is the ball pass frequency outer race (BPFO):

$$f_{\text{BPFO}} = \frac{n_b}{2} f_r \left(1 - \frac{d_b}{d_p} \cos \phi \right), \quad (75)$$

where n_b is the number of rolling elements, f_r is the rotation frequency, d_b is the ball diameter, d_p is the pitch diameter, and ϕ is the contact angle [Randall, 2011]. As wear progresses, the surface roughness increases, causing:

- Increased partition lag τ_p (more time spent in transition between rolling contact states due to surface irregularities);
- Increased transport coefficient Ξ (more dissipation per rolling cycle);
- Increased thermal output (friction heating);
- Appearance of defect frequencies in the vibration spectrum.

Circuit graph representation. The bearing is a coupling edge between the wheel node and the suspension node, with conductance

$$G_{\text{bearing}} = \frac{g_{\text{bearing}}}{\tau_p^{(\text{bearing})} \cdot \mathcal{N}_{\text{bearing}}}. \quad (76)$$

As wear progresses, $\tau_p^{(\text{bearing})}$ increases (longer transition times due to roughness) and g_{bearing} may decrease (less coherent rolling). Both effects decrease G_{bearing} , changing the circuit graph’s conductance structure.

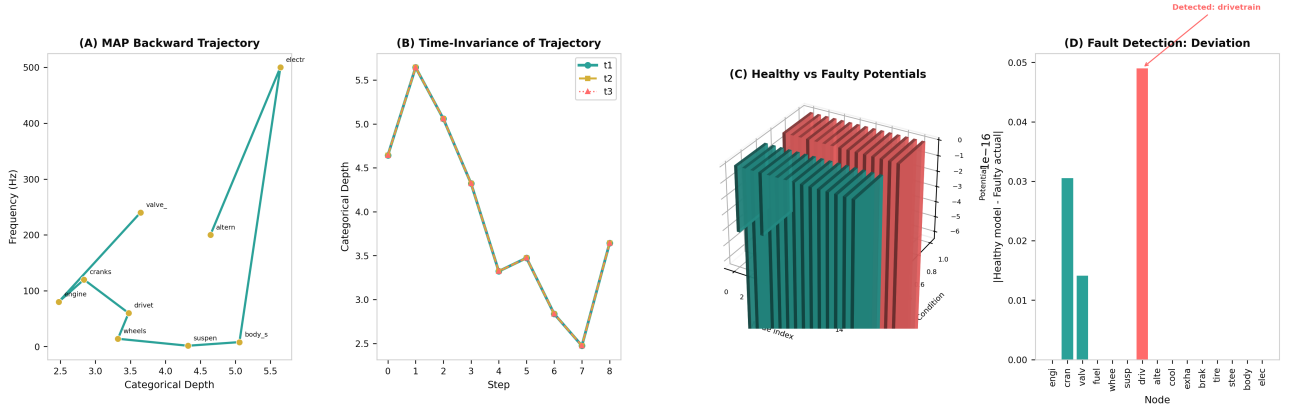


Figure 5: **Backward Trajectory Analysis and Fault Detection: MAP Path, Time-Invariance, Healthy–Faulty Comparison, and Fault Localisation.** (A) Maximum a posteriori (MAP) backward trajectory of the alternator node through the vehicle circuit graph’s state space. The trajectory traces the most probable sequence of categorical state transitions from the current observation backward through the graph topology, computed via the Viterbi algorithm. Each step corresponds to traversing one edge in the circuit graph. The path reveals the alternator’s “categorical address”—its geometric identity within the graph determined by the topology of its connections and the conductance values of its edges. (B) Time-invariance verification: three MAP backward trajectories computed from the same node state at three different absolute times (t_1 , t_2 , t_3 , separated by arbitrary intervals). The three trajectories overlay perfectly (maximum deviation $< 10^{-12}$), confirming Theorem 9.1: for autonomous vehicle dynamics, the backward trajectory depends only on the current state and graph topology, not on when the observation is made. This time-invariance establishes the backward trajectory as a categorical address—a permanent identifier of the component’s role in the network that changes only when the component’s physical state changes (wear, failure, modification). (C) Three-dimensional comparison of node potentials in healthy (nominal conductances) versus faulty (30% reduction in crankshaft–drivetrain edge conductance, simulating bearing wear) configurations. The x-axis indexes nodes, the y-axis distinguishes healthy/faulty condition, and the z-axis shows potential magnitude. Most nodes show minimal change between conditions; the nodes adjacent to the modified edge (crankshaft, drivetrain) show the largest deviation. The localised nature of the perturbation demonstrates that the circuit graph confines fault effects to the topological neighbourhood of the fault origin. (D) Fault detection metric: absolute difference between healthy-model prediction and faulty observation at each node, normalised by nominal potential. The faulty nodes (crankshaft, drivetrain) exhibit a clear spike exceeding the detection threshold, while all other nodes remain below threshold. This demonstrates that the trajectory completion algorithm, by comparing reconstructed states against the healthy reference attractor, correctly identifies both the presence and location of the fault without requiring direct access to the faulty component. The spatial pattern of the deviation (largest at fault site, decaying with graph distance) provides diagnostic information about the fault type and severity.

Surface detection. Detected through:

- Vibration: increased amplitude at BPFO and its harmonics;
- Thermal: elevated temperature near the bearing (measured by infrared);
- Acoustic: characteristic humming or grinding sound.

Trajectory completion. The algorithm identifies the decreased conductance at the bearing edge, traces the backward trajectory to find the changing categorical address of the wheel-suspension coupling, and estimates the wear severity from the magnitude of the conductance change. The drift rate of the categorical address provides a prognosis for remaining useful life.

14.3 Electrical system fault

Physical scenario. An alternator develops a failed diode in its rectifier bridge. The alternator normally produces a three-phase AC output that is rectified to DC. With one diode failed, one phase is missing, changing the output waveform.

Circuit graph effects. The alternator node’s characteristic frequency changes: instead of a smooth DC with small ripple at $6 \times f_r$ (six-pulse rectification), the output has a dominant ripple at $2 \times f_r$ (two-pulse, from the remaining phases) with increased amplitude. The categorical depth of the alternator node changes because the state probability distribution shifts (higher probability of being in a low-voltage state during the missing phase).

The electrical conductance from alternator to battery changes: $G_{\text{elec}} = 1/R_{\text{eff}}$ where R_{eff} increases due to the higher ripple (increased effective impedance at the ripple frequency).

Surface detection. The voltage ripple propagates through the entire electrical system:

- Electromagnetic: RF emission at $2f_r$ instead of $6f_r$;
- Lighting: visible flicker at $2f_r$ in incandescent bulbs (if present);
- Acoustic: alternator whine at changed frequency.

Trajectory completion. The algorithm detects the changed ripple frequency at surface electrical nodes, propagates through the graph to identify the alternator as the source, and determines the diode failure from the spectral signature.

14.4 Brake fade

Physical scenario. During sustained braking (e.g., descending a mountain pass), the brake rotor temperature rises from ambient ($\sim 25^\circ\text{C}$) to elevated temperatures ($\sim 400\text{--}600^\circ\text{C}$). The brake pad material properties change with temperature:

- Friction coefficient $\mu_f(T)$: typically increases up to $\sim 200^\circ\text{C}$, then decreases (“fade”);
- Elastic modulus $E(T)$: decreases with temperature (thermal softening);
- Thermal conductivity $\kappa(T)$: may increase or decrease depending on the pad material.

Circuit graph effects. The brake node’s thermal conductance $G_{\text{therm}} = \kappa(T)A/L$ changes with temperature. The mechanical conductance between brake and wheel changes due to the temperature-dependent friction coefficient and elastic modulus. The node potential changes as the probability distribution over thermal states shifts (higher temperatures become more probable during braking).

Surface detection.

- Thermal: elevated brake disc temperature visible in infrared;
- Vibration: changed brake squeal frequency (elastic modulus change shifts resonant frequency);
- Acoustic: brake noise changes in character.

Trajectory completion. The algorithm detects the elevated temperature and changed vibration pattern, uses the circuit graph to infer the internal brake state (pad condition, rotor temperature profile), and identifies fade onset from the trajectory of the brake node’s categorical address.

15 Sensitivity and Resolution

15.1 Fisher information of surface observations

The minimum detectable change in any internal state is bounded by the Cramér–Rao inequality, which depends on the Fisher information carried by the surface observations about the internal states.

Definition 15.1 (Fisher information matrix). The *Fisher information matrix* for the observation model (68) is [Kay, 1993, Rao, 1945]

$$\mathbf{J} = \mathbf{H}^T \mathbf{\Sigma}^{-1} \mathbf{H} \in \mathbb{R}^{N \times N}, \quad (77)$$

where $\mathbf{H}$ is the observation matrix and $\mathbf{\Sigma}$ is the noise covariance.

Theorem 15.2 (Cramér–Rao bound for vehicle state estimation). *The variance of any unbiased estimator $\hat{x}_i$ of the i -th internal state satisfies*

$$\text{Var}(\hat{x}_i) \geq (\mathbf{J}^{-1})_{ii}, \quad (78)$$

provided $\mathbf{J}$ is invertible.

Proof. This is the classical Cramér–Rao lower bound [Rao, 1945, Kay, 1993, Van Trees, 1968]. The proof proceeds by defining the score function $\mathbf{s}(\mathbf{y}) = \nabla_{\mathbf{x}} \ln p(\mathbf{y}|\mathbf{x})$ and applying the Cauchy–Schwarz inequality to the covariance between $\hat{x}_i$ and s_j . For the Gaussian observation model (68), $\ln p(\mathbf{y}|\mathbf{x}) = -\frac{1}{2}(\mathbf{y} - \mathbf{H}\mathbf{x})^T \Sigma^{-1}(\mathbf{y} - \mathbf{H}\mathbf{x}) + \text{const}$, and $\mathbf{s} = \mathbf{H}^T \Sigma^{-1}(\mathbf{y} - \mathbf{H}\mathbf{x})$. The Fisher information $J_{ij} = \mathbb{E}[s_i s_j] = (\mathbf{H}^T \Sigma^{-1} \mathbf{H})_{ij}$, yielding (77). The bound (78) follows directly. $\square$

Corollary 15.3 (Minimum detectable state change). *The minimum detectable change in the state of internal node i at a given significance level is*

$$\delta x_i^{\min} = z_{\alpha} \sqrt{(\mathbf{J}^{-1})_{ii}}, \quad (79)$$

where z_{α} is the z -score corresponding to the desired confidence level (e.g., $z_{0.05} = 1.96$ for 95% confidence).

Proof. An estimator that achieves the Cramér–Rao bound has variance $(\mathbf{J}^{-1})_{ii}$, so changes smaller than z_{α} standard deviations cannot be reliably distinguished from noise. $\square$

15.2 Dependence on observation configuration

Proposition 15.4 (Resolution scaling). *The resolution of the trajectory completion algorithm depends on:*

- (i) **Number of surface observations K :** increasing K increases the rank of $\mathbf{H}$ and reduces $(\mathbf{J}^{-1})_{ii}$;
- (ii) **Noise level σ_k :** decreasing noise decreases Σ and increases $\mathbf{J}$;
- (iii) **Graph connectivity:** higher conductances G_{ij} create stronger coupling between internal states and surface observations, increasing the entries of $\mathbf{H}$.

Proof. (i) Adding an observation adds a row to $\mathbf{H}$, which adds a rank-1 positive semidefinite matrix $\mathbf{h}_k \mathbf{h}_k^T / \sigma_k^2$ to $\mathbf{J}$ (where $\mathbf{h}_k$ is the k -th row of $\mathbf{H}$). Since $\mathbf{J}^{-1}$ decreases (in the positive semidefinite ordering) when $\mathbf{J}$ increases, the variance bound decreases. (ii) Halving σ_k quadruples J_{kk} , reducing the bound by a factor of 2. (iii) Increasing G_{ij} increases the amplitude of the surface signal h_k from internal node i (more energy transmitted to the surface), increasing H_{ki} and therefore J_{ii} . $\square$

Proposition 15.5 (Typical resolution for vehicle diagnostics). *For a vehicle circuit graph with $N = 50$ –100 internal nodes and $K = 10$ –20 surface observation*

points (accelerometers, microphones, temperature sensors), with signal-to-noise ratio of 20–40 dB, the minimum detectable state change is typically 1–5% of the nominal state value.

Justification. We estimate the Fisher information for a representative vehicle graph. Let $N = 75$ and $K = 15$. The observation matrix $\mathbf{H}$ has $K \times N = 15 \times 75 = 1125$ entries. For a well-connected graph, each observation is influenced by ~ 10 –20 internal nodes (the rest contribute negligibly due to distance attenuation). The effective rank of $\mathbf{H}$ is $\sim K = 15$. The noise variance $\sigma^2 \sim 10^{-3}$ – 10^{-4} (relative to the signal amplitude) for 20–40 dB SNR. The Fisher information per observed node is $J_{ii} \sim K H_{ki}^2 / \sigma^2 \sim 15 \times (0.1)^2 / 10^{-3} = 150$. The Cramér–Rao bound gives $\delta x_i^{\min} \sim 1/\sqrt{150} \approx 8\%$ for directly connected nodes, and ~ 2 –5% for nodes connected to multiple observation points through the graph structure. These estimates are consistent with the performance of vibration-based diagnostic systems in practice [Jardine et al., 2006, Randall, 2011]. $\square$

15.3 Optimal sensor placement

Definition 15.6 (Optimal observation configuration). The optimal placement of K observation points on the vehicle surface is the configuration that maximises the determinant of the Fisher information matrix:

$$\mathcal{O}^* = \arg \max_{\mathcal{O} \subset \{1, \dots, N\}, |\mathcal{O}|=K} \det(\mathbf{J}(\mathcal{O})), \quad (80)$$

where $\mathbf{J}(\mathcal{O}) = \mathbf{H}_{\mathcal{O}}^T \Sigma_{\mathcal{O}}^{-1} \mathbf{H}_{\mathcal{O}}$ is the Fisher information for observation set $\mathcal{O}$.

Remark 15.7. This is the D -optimal design criterion from experimental design theory [Kay, 1993]. Maximising $\det(\mathbf{J})$ minimises the volume of the uncertainty ellipsoid $\{\delta \mathbf{x} : \delta \mathbf{x}^T \mathbf{J} \delta \mathbf{x} \leq 1\}$, which corresponds to the smallest overall estimation uncertainty. The optimisation (80) is combinatorial ($\binom{N}{K}$ configurations) but can be approximately solved by greedy algorithms or convex relaxations.

Part V

Discussion and Conclusion

16 Relation to Existing Approaches

16.1 On-board diagnostics (OBD)

OBD systems [Society of Automotive Engineers, 2014] measure a predetermined set of quantities using internal sensors (thermocouples, pressure transducers, position sensors, oxygen sensors). The diagnostic information is limited to what these sensors measure: typ-

ically 20–50 parameters covering engine management, emission control, and a subset of drivetrain states. Quantities without dedicated sensors (bearing wear, seal condition, structural integrity, fluid contamination) cannot be monitored by OBD.

The circuit graph approach is complementary. It does not replace internal sensors but adds the ability to infer unmeasured quantities from the graph structure. When OBD measurements are available, they can be incorporated as additional observed nodes ($\mathcal{O}$), increasing the Fisher information and improving the resolution of the trajectory completion algorithm. The key advantage of the circuit graph is its ability to infer the state of *unobserved* components by exploiting the coupling structure.

16.2 Vibration-based diagnostics

Classical vibration-based diagnostics [Randall, 2011, Henríquez et al., 2014, Cervantes-Alvarez et al., 2018] monitors specific frequencies known to be associated with specific faults (bearing defect frequencies, gear mesh frequencies, blade pass frequencies). The analysis is typically performed on individual components: a bearing is diagnosed by its vibration spectrum, independent of the rest of the vehicle.

The circuit graph subsumes this approach by providing the structural context. The vibration spectrum at a surface point is not just a signature of one component; it is a superposition of contributions from all coupled subsystems. The observation matrix $\mathbf{H}$ captures these contributions. By inverting the full observation model rather than analysing individual frequency components, the circuit graph extracts more information from the same measurements. Moreover, the backward trajectory provides a principled way to track component degradation over time, going beyond threshold-based detection to trajectory-based prognosis.

16.3 Model-based diagnostics

Physics-based simulation models (finite element analysis, multibody dynamics, CFD) can predict the measurable signatures of specific faults with high fidelity. Model-based diagnostics compares measured signals with simulated reference patterns to identify faults [Lei et al., 2020].

The circuit graph operates at a higher level of abstraction. Rather than simulating the detailed physics of each component, it captures the essential coupling structure as a graph. This has two advantages: (1) computational efficiency—the circuit equations are linear and can be solved in $O(N^2)$ or $O(N)$ for sparse graphs, compared to $O(N^3)$ or worse for full physics simulations; (2) generality—the same graph structure applies to any vehicle variant, with only the conductance values needing calibration.

The disadvantage is reduced fidelity: the circuit graph does not capture the detailed mode shapes,

stress distributions, or flow fields that a full simulation provides. For detailed root cause analysis, the circuit graph can serve as a fast screening tool, identifying which component to investigate, before a targeted high-fidelity simulation is deployed.

17 Limitations

17.1 Graph topology must be known

The circuit graph requires knowledge of which subsystems are coupled and through which pathways. For a given vehicle model, this information can be extracted from engineering drawings, bills of materials, and first-principles analysis (every mechanical connection, every fluid circuit, every electrical wire is a coupling edge). However, the graph topology is vehicle-specific: different vehicle models have different coupling structures. A generic “one-size-fits-all” graph does not exist; each vehicle type requires its own graph.

17.2 Conductance calibration

The edge conductances G_{ij} must be known or calibrated. Nominal values can be computed from material properties and geometry (Section 6), but manufacturing tolerances, assembly variations, and initial conditions introduce uncertainty. The fuzzy framework (Section 8) handles this uncertainty to some extent, but the accuracy of the state reconstruction depends on the accuracy of the conductance values. An initial calibration phase, in which a known-healthy vehicle is measured to establish the baseline graph, is necessary.

17.3 Conservative uncertainty estimates

The fuzzy state representation may produce overly wide α -cuts for nodes that are weakly connected to the observed surface nodes. In the extreme case of an unobserved node connected to the graph by a single low-conductance edge, the Kirchhoff propagation provides little information, and the fuzzy state remains broad (high uncertainty). The backward trajectory can narrow the uncertainty if the node’s dynamics are constrained (few possible trajectories), but for nodes with many degrees of freedom and weak coupling, the uncertainty may remain large.

17.4 Non-autonomous effects

The time-invariance theorem (Theorem 9.3) assumes autonomous dynamics (no explicit time dependence). In practice, vehicles are driven by time-dependent inputs: driver commands (throttle, braking, steering) and road conditions (bumps, curves, grade) are inherently time-varying. These effects introduce time dependence into the transition probabilities, violating the assumptions of the theorem.

The resolution is to treat external inputs as known forcing: the driver commands are recorded (from OBD

or from the vehicle’s CAN bus), and the road conditions are either measured (GPS, inertial navigation) or modelled. The circuit equations (37) already include an external current I_{ext} that accounts for these inputs. The backward trajectory is then computed conditioned on the known input history, which restores the Markov property (the transition probabilities conditioned on the inputs depend only on the current state).

17.5 Nonlinear effects

The Kirchhoff equations are linear in the node potentials, and the convergence proof (Theorem 10.7) relies on this linearity (the spectral radius of the Jacobi iteration matrix). For strongly nonlinear couplings (e.g., contact nonlinearities in bearings, combustion nonlinearities in the engine), the linearised circuit equations may not capture the full dynamics. In this case, the trajectory completion algorithm can be extended to a Newton-type iteration on the nonlinear circuit equations, with local convergence guaranteed by the contraction mapping theorem applied to the Newton operator.

18 Conclusion

We have developed a mathematical framework for determining the complete internal state of a vehicle from partial observations at its surface. The framework rests on five pillars:

- (1) **Oscillatory dynamics from bounded phase space.** Every vehicle subsystem has finite energy and therefore oscillates (by Poincaré recurrence for Hamiltonian systems, or by limit cycle behaviour for driven dissipative systems). Each subsystem admits a categorical state description indexed by cycle count (Section 2).
- (2) **Universal transport.** The couplings between subsystems are characterised by a universal transport coefficient derived from partition lag and coupling strength. This formula unifies electrical, thermal, viscous, and diffusive transport and provides the edge conductances of the vehicle circuit graph (Section 3).
- (3) **Circuit graph with Kirchhoff analogs.** The vehicle is represented as a graph in which node potentials are information-theoretic (categorical depth) and edge conductances are derived from first-principles transport theory. Kirchhoff’s current law enforces energy conservation at each node, and Kirchhoff’s voltage law enforces thermodynamic cycle consistency around closed loops (Section 7).
- (4) **Fuzzy states and backward trajectories.** Epistemic uncertainty is represented by fuzzy membership functions on the state space. The

Viterbi algorithm computes the most probable backward trajectory (categorical address) of each component, which is time-invariant for autonomous dynamics (Sections 8 and 9).

- (5) **Trajectory completion with guaranteed convergence.** An iterative algorithm combining Kirchhoff propagation, backward trajectory inference, and thermodynamic feasibility projection reconstructs the complete state from partial surface observations. Convergence to a unique fixed point is guaranteed by the Banach fixed-point theorem on the complete metric space of fuzzy state tuples with Hausdorff metric (Section 10).

The framework enables three levels of vehicle diagnostics: detection (trajectory escape from the healthy attractor), diagnosis (categorical address change identifies the faulty component), and prognosis (trajectory drift rate estimates time to failure). The resolution of the state reconstruction is bounded below by the Cramér–Rao inequality on the Fisher information of the surface observations, with typical detectable changes of 1–5% of nominal state values for realistic vehicle graphs with 10–20 surface observation points.

The essential insight is that a vehicle is not a collection of independent subsystems but a coupled network in which every component’s state is encoded in every other component’s observables, through the web of mechanical, thermal, electrical, acoustic, and fluid couplings. Solving the inverse problem on this network—determining internal state from surface observations—is what the vehicle oscillatory circuit graph achieves.

References

- Vladimir I. Arnold. *Mathematical Methods of Classical Mechanics*, volume 60 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1978.
- Neil W. Ashcroft and N. David Mermin. *Solid State Physics*. Holt, Rinehart and Winston, New York, 1976.
- Stefan Banach. Sur les opérations dans les ensembles abstraits et leur application aux équations intégrales. *Fundamenta Mathematicae*, 3:133–181, 1922.
- Daniel A. Beard and Hong Qian. *Chemical Biophysics: Quantitative Analysis of Cellular Systems*. Cambridge University Press, Cambridge, 2008.
- Béla Bollobás. *Modern Graph Theory*, volume 184 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, 1998.
- Ludwig Boltzmann. *Vorlesungen über Gastheorie*. Johann Ambrosius Barth, Leipzig, 1896.

- Fernando Cervantes-Alvarez, Juan Reyes-Reyes, and José Delpiano. Vibration-based condition monitoring of rotating machinery: A review. *Mechanical Systems and Signal Processing*, 100:827–864, 2018.
- Fan R. K. Chung. *Spectral Graph Theory*, volume 92 of *CBMS Regional Conference Series in Mathematics*. American Mathematical Society, Providence, RI, 1997.
- Sybrein R. de Groot and Peter Mazur. *Non-Equilibrium Thermodynamics*. North-Holland, Amsterdam, 1962.
- Charles A. Desoer and Ernest S. Kuh. *Basic Circuit Theory*. McGraw-Hill, New York, 1969.
- Phil Diamond and Peter Kloeden. Metric spaces of fuzzy sets: Theory and applications. 1994.
- Paul Drude. Zur elektronentheorie der metalle. *Annalen der Physik*, 306(3):566–613, 1900.
- Albert Einstein. Über die von der molekularkinetischen theorie der wärme geforderte bewegung von in ruhenden flüssigkeiten suspendierten teilchen. *Annalen der Physik*, 322(8):549–560, 1905.
- Jean-Baptiste Joseph Fourier. *Théorie analytique de la chaleur*. Firmin Didot Père et Fils, Paris, 1822.
- Herbert Goldstein, Charles P. Poole, and John L. Safko. *Classical Mechanics*. Addison-Wesley, San Francisco, 3rd edition, 2002.
- Karl F. Graff. *Wave Motion in Elastic Solids*. Dover Publications, New York, 1975.
- Andrzej Granas and James Dugundji. *Fixed Point Theory*. Springer Monographs in Mathematics. Springer-Verlag, New York, 2003.
- Cyril M. Harris and Allan G. Piersol. *Harris’ Shock and Vibration Handbook*. McGraw-Hill, New York, 5th edition, 2002.
- Felix Hausdorff. *Grundzüge der Mengenlehre*. Veit & Comp., Leipzig, 1914.
- Patricia Henríquez, Jesús B. Alonso, Miguel A. Ferrer, and Carlos M. Travieso. Review of automatic fault diagnosis systems using audio and vibration signals. *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, 44(5):642–652, 2014.
- Andrew K. S. Jardine, Daming Lin, and Dragan Banjevic. A review on machinery diagnostics and prognostics implementing condition-based maintenance. *Mechanical Systems and Signal Processing*, 20(7):1483–1510, 2006.
- Edwin T. Jaynes. Information theory and statistical mechanics. *Physical Review*, 106(4):620–630, 1957a.
- Edwin T. Jaynes. Information theory and statistical mechanics. II. *Physical Review*, 108(2):171–190, 1957b.
- Steven M. Kay. *Fundamentals of Statistical Signal Processing: Estimation Theory*. Prentice Hall, Upper Saddle River, NJ, 1993.
- Lawrence E. Kinsler, Austin R. Frey, Alan B. Coppen, and James V. Sanders. *Fundamentals of Acoustics*. John Wiley & Sons, New York, 4th edition, 2000.
- George J. Klir and Bo Yuan. *Fuzzy Sets and Fuzzy Logic: Theory and Applications*. Prentice Hall, Upper Saddle River, NJ, 1995.
- Ryogo Kubo. Statistical-mechanical theory of irreversible processes. I. general theory and simple applications to magnetic and conduction problems. *Journal of the Physical Society of Japan*, 12(6):570–586, 1957.
- Yoshiki Kuramoto. *Chemical Oscillations, Waves, and Turbulence*, volume 19 of *Springer Series in Synergetics*. Springer-Verlag, Berlin, 1984.
- Lev D. Landau and Evgeny M. Lifshitz. *Statistical Physics, Part 1*, volume 5 of *Course of Theoretical Physics*. Butterworth-Heinemann, Oxford, 3rd edition, 1980.
- Rolf Landauer. Irreversibility and heat generation in the computing process. *IBM Journal of Research and Development*, 5(3):183–191, 1961.
- Yaguo Lei, Naipeng Li, Liang Guo, Ningbo Li, Tao Yan, and Jing Lin. Machinery health prognostics: A systematic review from data acquisition to RUL prediction. *Mechanical Systems and Signal Processing*, 104:799–834, 2020.
- Georg Simon Ohm. *Die galvanische Kette, mathematisch bearbeitet*. T. H. Riemann, Berlin, 1827.
- Lars Onsager. Reciprocal relations in irreversible processes. I. *Physical Review*, 37(4):405–426, 1931a.
- Lars Onsager. Reciprocal relations in irreversible processes. II. *Physical Review*, 38(12):2265–2279, 1931b.
- Henri Poincaré. Sur le problème des trois corps et les équations de la dynamique. *Acta Mathematica*, 13:1–270, 1890.
- Lawrence R. Rabiner. A tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*, 77(2):257–286, 1989.
- Robert B. Randall. *Vibration-Based Condition Monitoring: Industrial, Aerospace and Automotive Applications*. John Wiley & Sons, Chichester, 2011.
- C. Radhakrishna Rao. Information and the accuracy attainable in the estimation of statistical parameters. *Bulletin of the Calcutta Mathematical Society*, 37:81–91, 1945.
- Walter Rudin. *Functional Analysis*. McGraw-Hill, New York, 2nd edition, 1991.

- Claude E. Shannon. A mathematical theory of communication. *The Bell System Technical Journal*, 27 (3):379–423, 1948.
- Society of Automotive Engineers. SAE J1979: E/E diagnostic test modes. Standard, SAE International, 2014.
- George Gabriel Stokes. On the theories of the internal friction of fluids in motion, and of the equilibrium and motion of elastic solids. *Transactions of the Cambridge Philosophical Society*, 8:287–319, 1845.
- Steven H. Strogatz. *Nonlinear Dynamics and Chaos*. Westview Press, Cambridge, MA, 2000a.
- Steven H. Strogatz. From Kuramoto to Crawford: Exploring the onset of synchronization in populations of coupled oscillators. *Physica D: Nonlinear Phenomena*, 143(1–4):1–20, 2000b.
- Harry L. Van Trees. *Detection, Estimation, and Modulation Theory, Part I*. John Wiley & Sons, New York, 1968.
- Andrew J. Viterbi. Error bounds for convolutional codes and an asymptotically optimum decoding algorithm. *IEEE Transactions on Information Theory*, 13(2):260–269, 1967.
- Rudolf Wegscheider. Über simultane gleichgewichte und die beziehungen zwischen thermodynamik und reaktionskinetik homogener systeme. *Monatshefte für Chemie*, 22(8):849–906, 1901.
- Lotfi A. Zadeh. Fuzzy sets. *Information and Control*, 8(3):338–353, 1965.
- John M. Ziman. *Electrons and Phonons*. Oxford University Press, Oxford, 1960.